

The Reversal of the Gender Education Gap with Economic Development

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Abstract. The strikingly large gender education gap in low-income countries robustly narrows and reverses with economic development. To study the driving forces quantitatively, we propose a three-sector framework in which development features exogenous skill-biased technological change, structural change, gender-biased technological change, changes in marriage markets, and home productivity improvement. The model is parameterized to match contrasting labor market outcomes by education and gender across the development spectrum. Counterfactual exercises show that skill-biased technological change and structural change explain most of the narrowing gender education gap. Our model suggests that the marketization of services becomes important only after economics are sufficiently developed.

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1 Introduction

In almost all industrialized countries, economic development has been accompanied by dramatic progress in gender equality, most notably in educational attainment. To date, among 25- to 34-year-olds, women are more likely to have a tertiary degree than men in all 38 OECD countries (Goldin et al., 2006; Wolfers, 2021). In low-income countries, however, women are less than half as likely to have a tertiary degree or a secondary degree as men. As higher education among females is strongly correlated with female empowerment leading to more equitable and desirable gender outcomes¹, investigating the causes of cross-country differences in the gender gap in education is of paramount importance in understanding the progress of gender equality.

In this paper, we use nationally representative household surveys from 218 country-year samples covering countries of all income levels to document particularly large variations in the gender education gap across countries. We find that the gender education gap narrows sharply with economic development and reverses in many high-income countries. In addition, younger cohorts in high- and middle-income countries are making fast progress compared to older ones. However, in low-income countries, young cohorts are still only making slow progress and a big gender education gap remains: even for cohorts born in the 1990s, women are only 60 percent as likely as men to complete university education. By conducting a shift-share accounting exercise, we find that cross-country gender differences in the distributions of occupation and employment sector account for around 40 percent of the narrowing gender education gap with development, pointing to labor market outcomes as a crucial contributor to the observed education patterns.

Motivated by the empirical facts, this paper systematically investigates the effects of four mechanisms to understand the narrowing and reversal of the gender education gap with development: (i) simultaneous skill-biased technological change (SBTC) and structural transformation (ST), (ii) gender-biased technological change (GBTC), (iii) home technological changes (HTC), and (iv) changing marriage markets. The combination of SBTC and ST leads to a rise in skilled services in which women have a comparative advantage, thus increasing the relative labor demand of educated women. GBTC can be interpreted either as technological change or a reduced form of modeling a fall in gender discrimination, which changes women’s labor market opportunities across all sectors. HTC matters for women’s labor supply decisions and hence education decisions because home production is the op-

¹For example, Lise and Seitz (2011) show that increasing the education levels of women leads to a decrease in consumption inequality within households; Doepke and Tertilt (2018) show that the desired fertility of higher-educated women has a larger impact on realized fertility compared to that of less-educated women.

portunity cost of working in the market. Lastly, for the marriage market, the expectation of educational assortative matching shapes women’s incentives to obtain education even when working hours are low, as a result of skewed matching probabilities towards similarly educated men.

To formalize and quantify this idea, we develop a tractable general equilibrium model featuring the four mechanisms above.² Female and male agents are heterogeneous in their cost of obtaining education as in [Rendall \(2024\)](#). Agents make education choices to maximize expected utility. Then a male and a female mechanically form a household according to varying levels of educational assortative matching. The household jointly makes female labor force participation and consumption decisions to maximize the household utility subject to the budget constraint. Females who are not in the labor force produce only home goods by construction. Income is a function of the household education combination and the woman’s labor force participation — workers earn a gender- and education-specific wage in equilibrium.

On the production side, our model is built to analytically identify some key technological parameters. Specifically, there are three types of firms; they use technologies with potentially different skill and gender intensities to produce agriculture, manufacturing, and service goods. The production of each type of firm employs high- and low-educated labor as imperfect substitutes, whereas each education type of labor combines female and male labor as imperfect substitutes.

To assess the model’s quantitative predictions, we parameterize the model using moments in the U.S. data between 1980 and 2005, which is also the time frame with available data across a large sample of countries. We use model identities to directly identify production technologies governing GBTC and SBTC using data moments of the gender wage gap, labor force shares, wage bill shares of the skilled, and sectoral college premia. ST is directly determined by estimates of real sectoral productivity growth, and assortative matching parameters are calculated on the basis of the distribution of education combinations across marriages. We discipline the remaining three sets of parameters through a calibration: i) the distributions of the education cost of women and men are largely informed by the education shares by gender; ii) the home productivity distributions of the educated and uneducated women are informed by the female labor force participation rate (LFPR) by education; and iii) the consumption parameters are informed by the U.S. time series of sectoral value-added shares.

Note that we do not assume the direction of technological changes or in which sector(s) fe-

²We build the model based on the frameworks proposed by [Kongsamut et al. \(2001\)](#), [Ngai and Pissarides \(2007\)](#), [Ngai and Petrongolo \(2017\)](#), [Rendall \(2018\)](#), and [Buera et al. \(2021\)](#).

males have a comparative advantage; instead, we let the model identify the productivity and gender parameters using observable moments. Consistent with the literature, our parameterization suggests that i) females do have a comparative advantage in service production; ii) females on average have lower costs of education than males; and iii) firms face SBTC and ST simultaneously with economic development.

Given the parameterized model, we first conduct counterfactuals to quantify the contributions of different mechanisms in explaining the reversal of the gender education gap in the U.S. We find that, while SBTC alone explains one-fifth of the closing gap, it produces a fall in the LFPR of both educated and uneducated women, contradicting the data. This is because, given men’s comparative advantage in manufacturing and agriculture, SBTC in these two sectors decreases women’s incentive to enter the labor market and even widens the gender education gap, working against the effects of SBTC in services. ST alone can account for over half of the closing gap, but also falls short of producing a reversal. In contrast to the SBTC mechanism, ST incentivizes women to enter the labor market as the opportunity cost of staying at home increases with the growth of services, which affects both educated and uneducated women. However, we find it is crucial for SBTC, especially in services, and ST to work as complements to reverse the gender education gap in the U.S., with a joint explanatory power of around 117 percent. Meanwhile, GBTC contributes only a minor role (7 percent) to the observed closing gap, while a fall in assortative matching among the educated and an increase in home productivity attenuate the effects of SBTC and ST on reversing the gender education gap in general equilibrium.

We then validate the model by assessing its predictions of the gender education gap in other high-income countries between 1980 and 2005. To do so, we vary i) sectoral productivity and skill intensity, ii) gender-specific productivity parameters, which are directly identified from the data through model analytical results, across country-year samples, and iii) home productivity parameters which are disciplined by females’ LFPR, while fixing iv) the assortative matching parameter at the U.S. 2005 level due to a lack of data availability.³ We keep all the other parameters at the U.S. level in the corresponding year. We find that the model’s prediction of the gender education gap explains 86 percent of the variations in the data, providing a satisfactory fit. In particular, the combination of SBTC and ST explains all or most of the reversal of the gender education gap in Finland, Great Britain, Belgium, Japan, and Korea, but it explains none of it in Italy. Furthermore, the counterfactual exercises confirm that the complementarity between SBTC and ST remains the single most important driver in reversing the gender education gap — GBTC and changes in home production play only

³The results remain almost identical if we assume that the assortative matching parameter in these countries varies from 1980 to 2005 as in the U.S.

a minor role.

To understand cross-country patterns across the economic development spectrum, we take the benchmark parameterized model but estimate SBTC, ST, HTC, and varying assortative matching across countries. Specifically, we calibrate the sectoral productivity, skill intensity, and home productivity parameter values to closely match the key female labor market outcomes by income level, as the direct moments to estimate them are not available. The assortative matching parameters are directly determined by empirical estimates as before. We keep GBTC parameters at the U.S. level as they cannot be identified due to a lack of reliable data on gender wage gaps. Without targeting, our model slightly over-predicts the gender education gap by overestimating the share of educated workers in agriculture. However, it accurately captures the difference in the gender education gap between high- and low-income countries. We conclude that, across countries, the narrowing gender education gap is consistently a consequence of simultaneous SBTC and ST, while the other two mechanisms play a negligible role.

Before concluding, we examine the role of the endogenous marketization of home production in narrowing the gender education gap. We find that the change between middle- and high-income countries mirrors that of the U.S. over time, with a surge in the employment of unskilled women in services. However, this strong pattern of marketization is *not* observed when comparing low- and middle-income countries as the low-skilled females' LFPR drops. The lack of marketization limits the extent to which the gender education gap narrows in middle-income countries.

Related Literature. What are the forces driving the narrowing gender gap in education with economic development? Existing studies link women's rising education rates to fertility behaviors (Rios-Rull and Sanchez-Marcos, 2002), to divorce risks (Guvenen and Rendall, 2015), to technical change favoring women (Rendall, 2018), and to women's formal working hours (Karapanagiotis and Reimers, 2024). However, none of these studies provide a systematic cross-country analysis. This paper draws on new evidence and theory to understand the link between the gender education gap and economic development.

Our work contributes to the literature on cross-country patterns of gender gaps in labor market outcomes. For example, Antecol (2000) discusses the cross-country gender gap in LFPR. More recently, Bridgman et al. (2018) show that while the market hours of women increase strongly with GDP per capita, those of men decrease somewhat; Bento et al. (2021) study the gender gap in entrepreneurship; and Chiplunkar and Kleineberg (2022) investigate the role of gender barriers versus economic forces in driving the gender gaps in employment and wages. Our work complements this literature by being the first to document and study

the causes of the reversal of the gender education gap with economic development.

Our paper builds on the literature trying to understand the link between structural change and economic development (see, for example, [Herrendorf et al., 2014](#); [Duarte and Restuccia, 2010](#), among others). Of particular relevance to our work is [Buera et al. \(2021\)](#), who use a broad panel of advanced economies to study how a systematic shift in the economy to sectors that are intensive in high-skill labor can explain the college premium variations. [Porzio et al. \(2022\)](#) use birth-cohort data on agricultural employment to show that human capital growth in the 20th century contributed to the global structural transformation. [Ngai et al. \(2022\)](#) document the patterns of women’s and men’s working hours over 150 years in the U.S. and relate them to structural transformation; however, they do not touch on education either empirically or theoretically. In this paper, we document the strikingly large gender education gap in low-income countries and show that structural change is a crucial driver of the narrowing gender education gap across countries.

Lastly, our paper is broadly related to the growing literature drawing on detailed micro evidence to document and understand cross-country labor market outcomes. For example, [Bick et al. \(2018\)](#) use household surveys from 80 countries to document that hours worked are higher on average in poorer countries. A series of recent studies use surveys across countries to investigate cross-country patterns of levels of and transitions between self-employment, unemployment, and wage employment ([Feng et al., 2023](#); [Poschke, 2022](#); [Donovan et al., 2022](#)). However, none of these studies address the gender education gap.

2 Data and Empirical Findings

In this section, we present empirical findings using microdata from 83 countries covering all income levels. We first document a robust narrowing and reversal of the gender education gap with economic development, both across countries and within a country over time. We then show the relevant labor market outcomes and assortative matching patterns across countries and within the U.S. over time.

We draw on household censuses and surveys from *IPUMS International*, which include individual-level information on age, gender, educational attainment, employment status, industry, occupation, marital status, and spousal education. For consistency, we keep only samples from 1980 to 2017, which cover 218 country-year censuses and surveys across 83 countries. We also show U.S. time-series patterns using data from 1980 to 2015, which is the period when most of the structural transformation toward services occurred in the U.S. For

estimates of GDP per capita, we use output-side real GDP at chained PPPs in 2011 US\$ (rgdpo) from the Penn World Table version 9.1 (PWT 9.1).

We restrict our attention to prime-age (ages 25-54) women and men throughout. We exclude samples with missing information on educational attainment and those living in group quarters. We use sample weights whenever they are available.

2.1 The gender education gap

We define the gender gap in education as the ratio of the share of females who have acquired education to that of males. Figure 1 plots the gender gaps in university completion and secondary school completion against log GDP per capita in Panel (a) and (b), respectively. The two dashed vertical lines divide the countries into the bottom, middle, and top terciles of the world income distribution.⁴

Panel (a) of Figure 1 highlights that in the poorest economies, women are less than half as likely to obtain a college degree as their male counterparts. This gender education gap narrows quickly with development levels and reverses in richer economies, where women are more likely than men to have a college degree. Importantly, the cross-country patterns are broadly consistent with the U.S. time series data — the gender gap is large in 1980, but steadily decreases, reaches equality in the 2000s, and then reverses. Panel (b) of Figure 1 shows that the patterns of the gender gap in completing secondary school with development are similar to that in completing university.

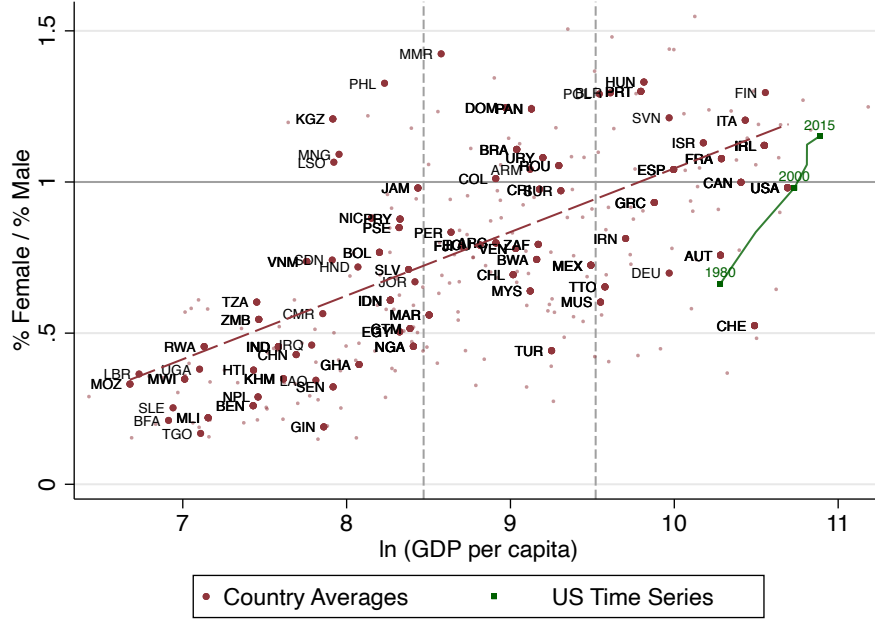
Table 1 reports the slope coefficients of regressing the ratio of women’s to men’s share of completing university and completing secondary school on log GDP per capita and a constant using different cuts of the data. When we include all 218 country-year samples separately for the gender gap in university completion, the slope coefficient is 0.20, which is close to the slope of using country averages at 0.21, as shown in Figure 1. We then consider the gender education gap in completing secondary school; the slope coefficients are 0.18 for both the country-year and country-average regressions. All four slopes are statistically significant at the one percent level.

We further explore how gender education gaps evolve over time within a country by comparing cross-sectional cohorts in all country-year samples. Figure 2 plots the gender education

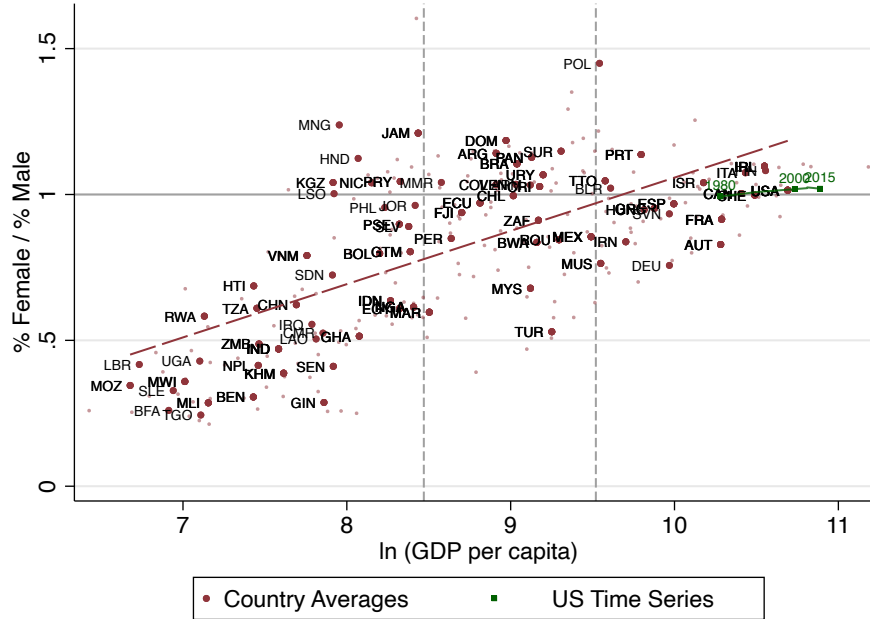
⁴Based on country-average GDP per capita from 1980 to 2017 for all countries in the PWT 9.1, the intervals for low- (bottom third of the world’s income distribution), middle- (middle third), and high-income (top third) countries are $\leq \$4,776$, between $\$4,776$ and $\$13,652$, and $\geq \$13,652$, respectively. These thresholds are close to the World Bank’s published thresholds for lower-middle-, upper-middle- and high-income countries.

Figure 1: Gender Gaps in Education

(a) University Completed



(b) Secondary School Completed



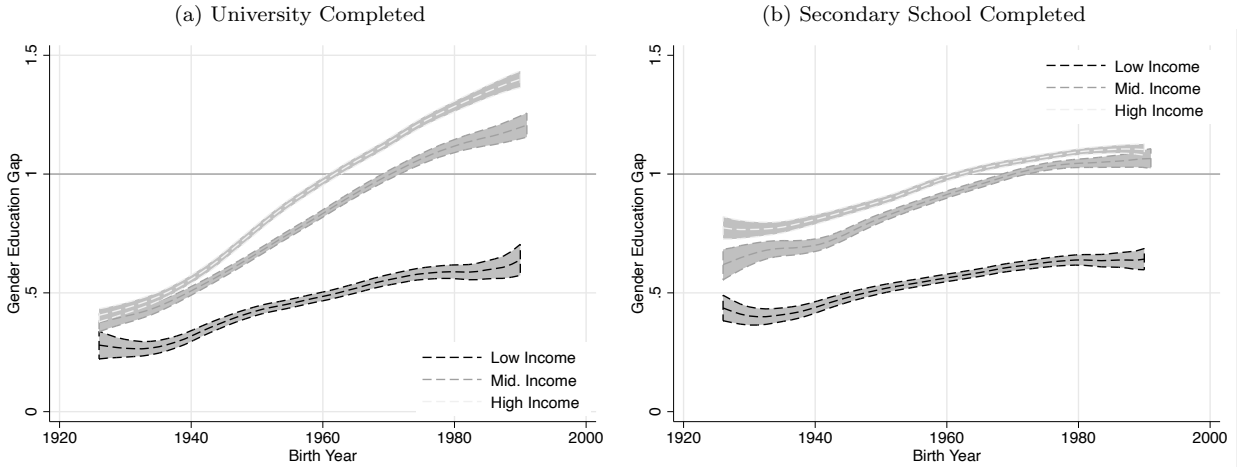
Note: This figure plots the female-to-male population shares of completing university and secondary school against log GDP per capita for each country-year sample (light red dot) and country averages (dark red dot). Red lines represent linear fitted lines of country averages. Green data points show the U.S. time trends.

Table 1: Regressions of the Gender Education Gap on the Development Level and a Constant

	By University		By Secondary	
	Country-year	Country-average	Country-year	Country-average
$\ln(\text{GDP per capita})$	0.20*** (0.014)	0.21*** (0.024)	0.18*** (0.011)	0.18*** (0.019)
R^2	0.41	0.44	0.46	0.46
Obs.	218	83	218	83

Note: The gender gap is defined as the share of females who complete university/secondary school divided by that of males. *** indicates statistical significance at the one-percent level. Standard errors are shown in parentheses.

Figure 2: Gender Education Gap by Cohort



Note: This figure plots the average gender education gap with the 95% confidence intervals by birth-year cohorts in low-, middle-, and high-income countries.

gap by birth-year cohorts in low-, middle-, and high-income countries. In middle- and high-income economies, gender education gaps reverse starting from cohorts born in the 1970s and 1960s, respectively. The low-income economies, however, still lag far behind. While the gender education gap has narrowed, even for cohorts born in the 1990s, females are only about 60 percent as likely as males to complete college or secondary school education.

The empirical results for university completion and secondary school completion are similar. Hereafter, we report the results for university completion as the education threshold in the

benchmark analysis.

Composition Effects. How much of the narrowing and reversal of the gender education gap with development can be accounted for by cross-country gender differences in demographic and labor-market compositions? To answer this question, we conduct a shift-share accounting exercise to find the composition effects of age, sector, and occupation using 204 country-year surveys with available information.⁵

Consider the share of educated females or males for country c in year t , $E_{ct}^g, g = f, m$, which can be written as the weighted average of the educated share across demographic groups $d \in D$ such that $E_{ct}^g = \sum_{d \in D} w_{dct}^g E_{dct}^g$, where w_{dct}^g is the population share of group d for gender g and E_{dct}^g is the corresponding educated share. To estimate the composition effects resulting from variations in w_{dct}^g , we construct a counterfactual aggregate educated share for each country-year observation by gender, $\tilde{E}_{ct}^g$, using the fixed weights $\bar{w}_{dct}^g$ of the global average across all country-year samples. Thus the counterfactual educated share is calculated as $\tilde{E}_{ct}^g = \sum_{d \in D} \bar{w}_{dct}^g E_{dct}^g$. Recall that our benchmark analysis regresses the aggregate gender education gap on log GDP per capita and a constant, $Gap_{ct} \equiv E_{ct}^f / E_{ct}^m = \alpha + \beta \log(y_{ct}) + \epsilon_{ct}$. We can now instead regress the counterfactual gender education gap $\tilde{Gap}_{ct}$ on GDP per capita and a constant, $\tilde{Gap}_{ct} \equiv \tilde{E}_{ct}^f / \tilde{E}_{ct}^m = \tilde{\alpha} + \tilde{\beta} \log(y_{ct}) + \tilde{\epsilon}_{ct}$. As a result, the composition effect of a set of demographic factors is given by the relative change in the estimates between the benchmark and counterfactual regressions, $1 - \frac{\tilde{\beta}}{\beta}$.

Table 2: Demographic Composition Accounting for the Aggregate Gender Education Gap

Factor	Age	Sector	Occupation
Accounting Share (%)	-6	30	43
Factors	Sector $\times$ Occupation Age $\times$ Sector $\times$ Occupation		
Accounting Share (%)	44 40		

Note: This table reports the percentage of the coefficient of regressing the gender education gap on log GDP per capita and a constant that is explained by cross-country gender differences in distributions of age, employment sector, occupation, and combinations of the above three factors using a shift-share accounting decomposition exercise.

As shown in Table 2, fixing the age distribution across countries slightly decreases the slope coefficient, while fixing the sector and occupation distributions have sizable explanatory power of 30 percent and 43 percent, respectively. The decomposition exercise points to labor

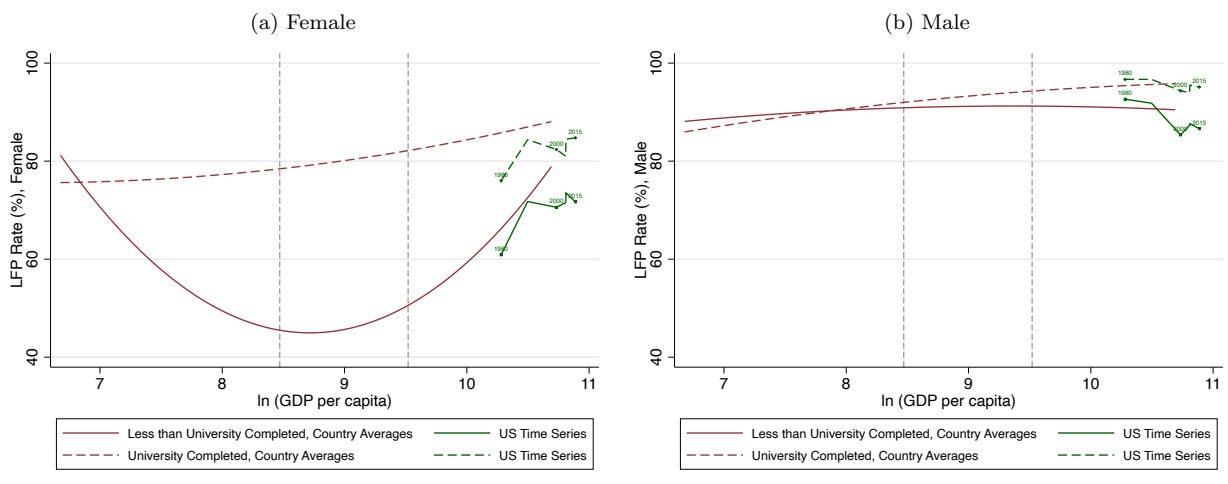
⁵Age is categorized into five-year age bins from 25 to 54; *Sector* includes agriculture, manufacturing, service, and the null; *Occupation* uses ISCO code and includes 10 categories and the null.

market outcomes as a potentially strong driving force behind the narrowing gender education gap across countries.

2.2 Labor market outcomes by gender and education

To motivate our modeling choice of SBTC, ST, and GBTC, we now turn to indicators of the labor market by gender and education attainment. The well-known U-shape aggregate female LFPR is dominated by the low-educated, resulting in an average LFPR of 56 percent, 49 percent, and 66 percent, respectively, for countries with available data in the world income distribution’s bottom, middle, and top terciles. Panel (a) of Figure 3 shows that the LFPR of women who have completed university *increases* with development and remains high at around 70–80 percent, while the LFPR of women who did not complete university is *U-shaped*. In contrast, men’s LFPR is always high for both education groups throughout different development levels, as shown in panel (b) of Figure 3. The aggregate males’ LFPR is 90 percent, 91 percent, and 92 percent, respectively, for the three terciles of the world income distribution.

Figure 3: Labor Force Participation Rate (%) by Gender and Education



Note: Red lines represent quadratic fitted lines of country-average labor force participation rates against log GDP per capita by gender and education level. We use the employment status variable (*empstat*) to classify all employed and unemployed individuals as being in the labor force, and all others as not being in the labor force.

Figure 3 also compares the U.S. time-series trend (in green) with the cross-country results (in red). Unlike the experience of the U.S. economy, when we compare low- and middle-income countries, the female LFPR decreases as the gender education gap narrows. Meanwhile,

consistent with the U.S., the female LFPR increases as the gender education gap continues to narrow and reverse between middle- and high-income countries. Therefore, it is crucial to understand whether the same mechanisms drive these different patterns of female LFPR with economic development and gender education gaps.

To further understand the link between the rise in women’s educational attainment and structural transformation, we investigate, both in the aggregate and by education levels, the sectoral female labor force share (FLFS) as a part of the total sectoral female and male labor. As the development level increases across countries, FLFS strongly decreases in the agricultural sector, increases in services with women’s labor share surpassing that of men, and remains the lowest and relatively stable in manufacturing. Furthermore, the sectoral FLFS patterns are completely driven by the low educated. Among the educated labor force, in contrast, FLFS is increasing in log GDP per capita for all three sectors, with the largest magnitude observed in services.⁶

To sum up, this section describes the patterns of labor force participation by gender and education, the rise of women’s employment in the service sector, and the rise of educated women’s employment in all sectors with development. These facts highlight the importance of considering SBTC and ST *simultaneously* in understanding the narrowing education gap.

2.3 Assortative household formation by education

Education decisions are influenced by not only labor market outcomes but also by household formation expectations. If educated individuals are more likely to marry educated and thus higher-income spouses, then education decisions are affected by the extent of assortative matching. In this section, we document the empirical regularities of assortative matching across development levels to mechanically discipline our quantitative model.

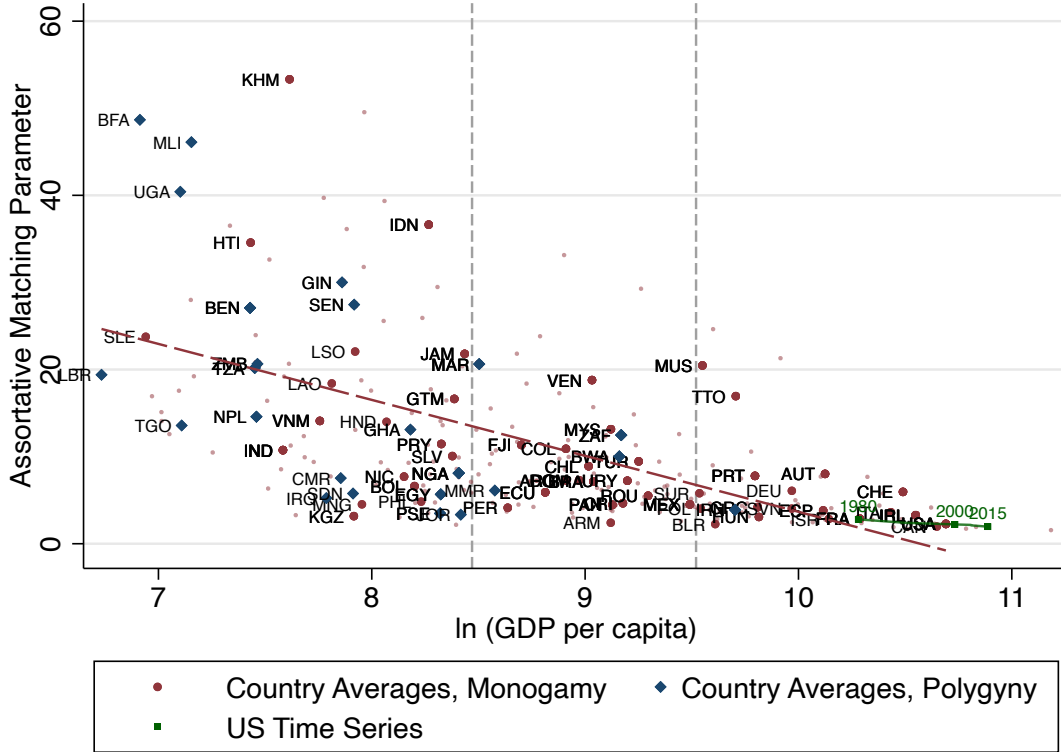
We follow Eika et al. (2019) to measure the educational assortative matching parameter as the probability ratio of the likelihood of an educated individual marrying an educated spouse relative to the likelihood of the pairing if matches were random. Suppose an economy consists of married households indicated by (j, j') , with $j, j' \in \{E, N\}$ representing the male’s and female’s education, respectively. λ_{ef} and λ_{em} denote the share of educated females and males, respectively. Under the assumption of random matching, the share of households where both the female and the male are educated is given by $\lambda_{ef}\lambda_{em}$. To mechanically account for the distribution of household education types, we denote by α the assortative

⁶Appendix Table A1 reports the slope coefficients of regressing sectoral LFLS on log GDP per capita and a constant in the aggregate and by education levels.

matching parameter such that the observed share of households where both the female and the male are educated is $\alpha\lambda_{ef}\lambda_{em}$. Using the accounting identities, we have the shares of households with education levels of $\{EE, EN, NE, NN\}$ given by

$$\begin{aligned}\%(EE) &= \alpha\lambda_{ef}\lambda_{em}, \\ \%(EN) &= \lambda_{em}(1 - \lambda_{ef}) + (1 - \alpha)\lambda_{em}\lambda_{ef}, \\ \%(NE) &= (1 - \lambda_{em})\lambda_{ef} + (1 - \alpha)\lambda_{em}\lambda_{ef}, \\ \%(NN) &= (1 - \lambda_{em})(1 - \lambda_{ef}) + (\alpha - 1)\lambda_{em}\lambda_{ef}.\end{aligned}$$

Figure 4: Educational Assortative Matching



Note: This figure plots α values against log GDP per capita for each country-year sample (light red dot) and country averages (blue diamond for countries that have polygamous unions and dark red dot for the rest). Observations with α values larger than the country-average 95th percentile value of 53.32 are dropped in this figure.

Consistent with Eika et al. (2019), we find that the U.S. assortative matching parameter α on university decreases from 2.80 in 1980 to 1.96 in 2015. Although we will only use α for educated workers to discipline household formation in the benchmark model, we can test whether our structural model produces consistent values of the assortative matching

parameter for the uneducated α_N . Denote by $\lambda_{nf}, \lambda_{nm}$ the share of uneducated females and males, respectively. By construction, $\alpha_N \equiv \frac{\%NN}{\lambda_{nf}\lambda_{nm}} = \frac{(1-\lambda_{ef})(1-\lambda_{em})}{\lambda_{nf}\lambda_{nm}} + \frac{(\alpha-1)\lambda_{ef}\lambda_{em}}{\lambda_{nf}\lambda_{nm}}$, which is pinned down by α and the education shares by gender. We will assess whether our model's prediction on α_N is consistent with the data in the quantitative analysis.

We further document that, across countries, the qualitative pattern of α decreasing with development also holds while the magnitudes become much larger. Figure 4 plots the benchmark estimates of α in all country-year samples when using university completion as the education threshold. The median value of α decreases from 16.6 in low-income countries to 4.0 in high-income countries for university-educated households. We also show in Appendix Figure A2 that the empirical assortative matching between the *uneducated* women and men marginally increases with development, consistent with the time-series patterns of high-income countries documented in Eika et al. (2019). We will incorporate the large variation in α across development levels in the model to investigate its role in explaining the narrowing gender education gap with development.⁷

Intuitively, since the degree of assortative matching among educated men and women decreases with economic development, this channel alone would decrease women's incentives to obtain education, especially when market hours are low. As a result, the gender education gap could potentially widen rather than narrow.

3 The model

This section introduces our benchmark general equilibrium model with endogenous education and female labor supply decisions. In our setup, large agricultural sectors in poor economies can be generated by either differential productivity growth rates by sector or non-homothetic preferences.

3.1 Households

There is a mass one of both females and males, heterogeneous in their cost of education. Individuals face a discrete education choice: to obtain an education or not. Those choosing an education must pay a utility cost, $a \in \mathbb{R}_+$, which follows the distribution $\Gamma(a)$. Individuals

⁷In the benchmark results, we follow the weighting mechanism in Eika et al. (2019) to include the single households when calculating the share of households where both the male and female are educated. Appendix Figure A3 plots the results when we restrict the sample to only married households. The patterns remain similar to our benchmark results.

also face a discrete choice in the labor market: to work in the agricultural sector, A , the manufacturing sector, M , or the service sector, S . Individuals, if working, supply one unit of labor to the market. Let $\mathbb{O} = \{NA, EA, NM, EM, NS, ES\}$ describe the set of education-sector possibilities. We assume free mobility across sectors; hence, wages by gender and education are equal across sectors.

Individuals make education decisions to maximize expected utility. After completing education, households composed of one male and one female are formed according to a mechanical assortative matching process. Households then make consumption and labor supply decisions to maximize utility. Denote the utility function of the household (j, j') by $u_{jj'}(c)$, where total consumption, c , consists of both home consumption, c_H , and market consumption.

Education decisions. For gender $g = f, m$, the expected utilities of completing and not completing education are $\mathbb{E}_E^g(u_{jj'}(c)) - a$ and $\mathbb{E}_N^g(u_{jj'}(c))$, respectively. Hence, an individual with an education cost of a will obtain education if and only if $a < \mathbb{E}_E^g(u_{jj'}(c)) - \mathbb{E}_N^g(u_{jj'}(c)) \equiv a^{*g}$. Therefore, females and males choose to acquire education if $a < a^{*g}$. The educated shares of females and males are $\Gamma(a^{*f})$ and $\Gamma(a^{*m})$, respectively.

Consumption and labor supply. We assume males inelastically supply one unit of labor. Home production is performed only by women and it takes a value drawn from the distribution $\ln(x) \sim \mathcal{N}_j(Z_h\mu_j, \sigma_j^2)$, $j = E, N$, where Z_h denotes the aggregate home productivity. The household consumes home-production output $c_H = x$ if the female does not work in the labor market, but consumes $c_H = \psi_j x$ if she works in the labor market, where $\psi_j < 1$, $j = E, N$. We assume that home production is only substitutable to services as commonly assumed in the literature.⁸ Let the composite of service consumption be

$$\hat{c}_S = \left(c_H^{\frac{\sigma-1}{\sigma}} + c_S^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \quad (1)$$

where c_S is the consumption of market services, and σ governs the elasticity of substitution between market and home-produced services. The household's total consumption is a constant-elasticity-of-substitution (CES) function of

$$c = \left(\phi_A^{\frac{1}{\nu}} (c_A - \underline{A})^{\frac{\nu-1}{\nu}} + \phi_M^{\frac{1}{\nu}} c_M^{\frac{\nu-1}{\nu}} + \phi_S^{\frac{1}{\nu}} \hat{c}_S^{\frac{\nu-1}{\nu}} \right)^{\frac{\nu}{\nu-1}}, \quad (2)$$

where c_A , c_M are agricultural and manufacturing sector-specific consumption, $\underline{A}$ is a constant subsistence level of consumption, and ν is the elasticity of substitution between consumption of all sectors.

⁸With technological progress, home production of agricultural and manufacturing goods disappears and only home production of services remains (Ngai and Pissarides, 2008).

Denote the female's labor supply dummy by $l_{jj'}^f$. Let $l_{jj'}^f = 1$ if the woman works and zero otherwise. We set $W = l_{jj'}^f$ to be the index of whether the female works in the market for a household type (j, j') . Hence, there are eight types of households characterized by $\{jj'W\}$ in equilibrium. Households (j, j') maximize,

$$\max_{c_A, c_M, c_S, c_H(l_{jj'}^f)} u_{jj'}(c) \quad (3)$$

s.t.

$$\sum_{i=A, M, S} p_i c_i \leq w_j^m + w_{j'}^f l_{jj'}^f \equiv y_{jj'W}, \quad (4)$$

where p_i denotes the output price of sector $i = A, M, S$, $w_{j'}^f$ and w_j^m denote the wage income for the female and male, respectively.

Since we consider only the extensive margin of labor market participation, without loss of generality, we focus on the case where home and market services are perfect substitutes $\sigma \rightarrow \infty$ or equivalently $\hat{c}_S = (c_H + c_S)$, which allows us to obtain the closed form solution for all household decisions. The main effect of imperfect substitution is that a woman's labor force participation will also depend on her husband's wage. Given that we have two types of husbands, educated or uneducated, with two distinct incomes, allowing for substitutability has little effect on our benchmark calibration results.⁹

Given market income $(y_{jj'W})$ and home production (c_H) , households maximise total consumption c . Let $\Omega = \phi_A p_A^{1-\nu} + \phi_M p_M^{1-\nu} + \phi_S p_S^{1-\nu}$, which we can interpret as the aggregate price index of the consumer. As shown in Appendix B.1, we can solve for the market consumption as

$$c_{jj'W} = (y_{jj'W} - p_A \underline{A} + p_S c_H) (\Omega)^{\frac{1}{\nu-1}}.^{10} \quad (5)$$

Therefore, a female enters the labor market if

$$u(c|c_H = \psi_j x) \geq u(c|c_H = x) \Rightarrow w_j^f \geq p_S x (1 - \psi_j) \equiv \underline{w}_j^f(x). \quad (6)$$

That is, market income needs to be large enough to compensate for any lost value-add of home production. A female also works if the male's wage does not cover the subsistence requirement in agricultural goods:

$$w_j^m < p_A \underline{A}. \quad (7)$$

⁹In the literature, estimates for the elasticity typically range between 1.8 and 2.5.

¹⁰In Appendix B.2, we also show the expression of consumption when there is imperfect substitution between market and home production.

As a result, women's employment choices are independent of male wages, as long as the spousal male's wage exceeds the subsistence level.

We denote $\underline{x}_j^f$ as the cutoff of home production such that $w_j^f(\underline{x}_j^f) = w_j^f$. In the case of households below the subsistence level, i.e., $w_j^m < p_A \underline{A}$, we set $\underline{x}_j^f \rightarrow +\infty$. Therefore, the shares of women in the labor force by education level are given by

$$L_E^f = \mathcal{N}_E(\underline{x}_E^f) \Gamma(a^{*f}), \quad (8)$$

$$L_N^f = \mathcal{N}_N(\underline{x}_N^f) (1 - \Gamma(a^{*f})). \quad (9)$$

3.2 Firms' production

There are three types of representative firms hiring labor to produce agriculture, manufacturing, and service outputs. The production functions for all three sectors are

$$Y_k = Z_k H_k, \text{ for } k = A, M, S.$$

The total sectoral labor input, H_k , is a CES function of educated and uneducated labor:

$$H_k = \left(\chi_k H_{Ek}^{\frac{\theta-1}{\theta}} + (1 - \chi_k) H_{Nk}^{\frac{\theta-1}{\theta}} \right)^{\frac{\theta}{\theta-1}}, \text{ for } k = A, M, S,$$

where θ is the elasticity between educated labor (H_{Ek}) and uneducated labor (H_{Nk}). Educated and uneducated labor inputs are CES functions of female labor (H_{Ek}^f) and male labor (H_{Ek}^m),

$$H_{jk} = F(H_{jk}^f, H_{jk}^m) = \left(\zeta_{jk} (H_{jk}^f)^{\frac{\eta-1}{\eta}} + (1 - \zeta_{jk}) (H_{jk}^m)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \text{ for } j = E, N,$$

where η is the elasticity between female and male labor.

We denote total labor force participation for gender g and education j by $H_j^g = \sum_k H_{jk}^g$. Because labor is perfectly mobile across sectors but not education levels, in equilibrium $w_{Ek}^f = w_E^f$, $w_{Nk}^f = w_N^f$, $w_{Ek}^m = w_E^m$, and $w_{Nk}^m = w_N^m$. Firms choose labor inputs to maximize profit:

$$\max_{\{H_{jk}^g\}} p_k Y_k - \sum_{jg} w_j^g H_{jk}^g.$$

Labors are paid with their marginal products, hence it is straightforward to solve the gender

wage gaps by education level:

$$x_j = \frac{w_j^f}{w_j^m} = \frac{\zeta_{jk}}{1 - \zeta_{jk}} \left(\frac{H_{jk}^f}{H_{jk}^m} \right)^{-\frac{1}{\eta}}, j = E, N. \quad (10)$$

Therefore, for a given η , rearranging equation (10) solves ζ_{jk} as an explicit function of directly observable data moments:

$$\zeta_{jk} = \frac{x_j \left(\frac{H_{jk}^f}{H_{jk}^m} \right)^{\frac{1}{\eta}}}{\left(1 + x_j \left(\frac{H_{jk}^f}{H_{jk}^m} \right)^{\frac{1}{\eta}} \right)}. \quad (11)$$

Now define p_{Ek} and p_{Nk} as the education-specific prices. Then the sectoral college wage premium, or the sector-specific relative price of college to non-college labor is, $\pi_k \equiv \frac{p_{Ek}}{p_{Nk}}$. Although π_k is not directly observed in the data, we use firms' zero-profit condition to solve the prices as a composite of wages by education and gender:

$$\pi_k \equiv \frac{p_{Ek}}{p_{Nk}} = \frac{\left[\zeta_{Ek}^\eta (w_E^f)^{1-\eta} + (1 - \zeta_{Ek})^\eta (w_E^m)^{1-\eta} \right]^{\frac{1}{1-\eta}}}{\left[\zeta_{Nk}^\eta (w_N^f)^{1-\eta} + (1 - \zeta_{Nk})^\eta (w_N^m)^{1-\eta} \right]^{\frac{1}{1-\eta}}}, k = A, M, S. \quad (12)$$

Hence, given ζ_{jk} and η , π_k can be directly computed by Equation (12) using data of hourly wage rate by gender and education. In addition, the sectoral educated wage bill share $I_{Ek} \equiv \frac{p_{Ek} H_{Ek}}{p_{Ek} H_{Ek} + p_{Nk} H_{Nk}}$ is directly available in the KLEMS dataset.

Finally, given π_k and I_{Ek} , we can solve for χ_k (see Appendix B.3 for details),

$$\chi_k = \frac{1}{1 + (\pi_k)^{\frac{1-\theta}{\theta}} (I_{Ek}^{-1} - 1)^{\frac{1}{\theta}}}, \text{ with } k = A, M, S. \quad (13)$$

3.3 Equilibrium and model predictions

A competitive equilibrium is defined by education choices (a^{*m}, a^{*f}) , market wages $(w_N^m, w_E^m, w_N^f, w_E^f)$, market prices (p_A, p_M, p_S) , consumption $\{c_{jj'WA}, c_{jj'WM}, c_{jj'WS}, c_{jj'WH}\}_{j,j'=E,N;W=0,1}$, and labor allocation $\{H_{Ek}^g\}_{g=f,m;k=A,M,S}$ and $\{H_{Nk}^g\}_{g=f,m;k=A,M,S}$, such that:

- (i) individuals make optimal education choices where men with $a < a^{*m}$ and women with $a < a^{*f}$ obtain an education, while others do not; females make optimal labor force participation choices where they work if $x > \underline{x}_j^f, j = E, N$.
- (ii) the representative firms maximize profits, subject to technology in Section 3.2; individ-

uals maximize utility (3), subject to the budget constraint (4);

(iii) given the optimal choices of firms and households, output prices clear the goods market in each sector, and market wages clear the labor market for each education-gender group:

$$\sum_{jj'=EE,EN,NE,NN;W=0,1} c_{jj'Wk} = Y_k, k = A, M, S; \quad (14)$$

$$H_{EA}^g + H_{EG}^g + H_{ES}^g + NL_E^g = \Gamma(a^{*g}), g = f, m; \quad (15)$$

$$H_{NA}^g + H_{NG}^g + H_{NS}^g + NL_N^g = 1 - \Gamma(a^{*g}), g = f, m. \quad (16)$$

Model predictions. There are four mechanisms that qualitatively affect the gender education gap in our model. The first mechanism is a combination of skill-biased technological change (SBTC) within sectors and structural transformation (ST) across sectors. SBTC is captured by increasing $\chi_k, k = A, M, S$, and ST arises due to a higher growth rate of Z_A, Z_M compared to Z_S . Consider the empirically relevant case where females have a comparative advantage in producing services, $\zeta_{jS} > \zeta_{jG}, \zeta_{jB}, j = E, N$, which will be confirmed later in our model calibration section. Disproportionately more females than males are drawn into the growing service sector due to ST. Furthermore, the increase in females' employment in the labor market incentivizes females to obtain more education due to SBTC, which narrows the gender education gap with development.

The second mechanism considers the gender-biased technological change (GBTC) captured by changes in $\zeta_{jk}, j = E, N, k = A, M, S$. One can interpret increasing ζ 's with economic development either as a technological change that makes female labor more productive or as a decline in the barriers or discrimination that women face in the labor market. This mechanism draws more women to market work, thus increasing women's incentive to obtain more education.

The third mechanism of home technological change (HTC) is driven by rising Z_h , which increases women's opportunity cost of working. This mechanism alone tends to decrease the incentives to obtain more education as education is usually rewarded more in the labor market.

The last mechanism concerns varying educational assortative matching. Our model predicts that an increase in assortative matching narrows the gender education gap. When α increases, educated females benefit from a higher probability of marrying an educated and thus higher-earning spouse. Meanwhile, the net benefits for educated males are lower; their utility increases with the educated spouse's higher income but decreases as the educated spouse is likely to opt out of full-time home production. Given the empirical observation of

α decreasing with development, our model predicts that variations in assortative matching actually widen the gender education gap in favor of men with development as a counter-force.

4 Quantitative Analysis

We investigate quantitatively the extent to which our model can account for the patterns of the gender education gap we document. In this section, we parameterize the model using the key features of the U.S. time series data. Using the parameterized model, we then conduct counterfactual exercises to quantify the role of each mechanism and their interactions in reversing the gender education gap in the U.S. Furthermore, we assess the model’s performance in predicting the variations in the gender education gap over time across high-income economies from the World KLEMS data.

4.1 Parameterization

We parameterize the model to replicate the U.S. economy in 1980 and in 2005, respectively.¹¹ In the benchmark, all preference and distribution parameters are imposed to be the same in the two equilibria, while the degree of SBTC, ST, GBTC, HTC, and assortative matching vary across time.

We start by normalizing TFP in the three market sectors to be one in 1980. We normalize the home productivity in 1980 to be 1 and set that in 2005 to be 1.10 following the hourly estimates in [Bridgman et al. \(2018\)](#). Home productivity differences between working and non-working women, $\psi_j, j = E, N$, are set to differ by home production hours. Using data from [Ramey and Ramey \(2010\)](#), we find that among educated females, those who are working spend 47 percent as much time in home production as those who are not working; among uneducated females, the corresponding figure is 54 percent. Hence, we set $\psi_E = 0.47$ and $\psi_N = 0.54$.

Next, we set all elasticities outside the model following the literature. The elasticity of substitution between sector-specific consumption, ν , has been estimated in numerous studies. For the benchmark calibration, we follow the recent estimates (see, for example, [Herrendorf et al., 2014](#)) using consumption value-added data and relative prices, which suggest a relatively low value of 0.002. The elasticity of substitution between educated and uneducated labor, θ , is set to 1.53 following [Buera et al. \(2021\)](#). This value is close to the estimates

¹¹We choose 2005 as the terminal year following [Buera et al. \(2021\)](#) because this is the last period that is consistently available across KLEMS datasets.

in the earlier literature. The elasticity of substitution between female and male labor, η , is set to 2.27, following [Ngai and Petrongolo \(2017\)](#) who match the response in hours ratios to changes in the wage ratio.

Table 3: Benchmark Parameters Values

Parameter	Value
Panel A: Pre-Assigned Parameters	
$\{Z_A^{US}, Z_M^{US}, Z_S^{US}\}_{t=0}$ - Sector-specific technology	$\{1, 1, 1\}$
$\{Z_h^{US}\}_{t=0,T}$ - Home productivity	$\{1, 1.10\}$
$\{\psi_E, \psi_N\}$ - Relative home productivity	$\{0.47, 0.54\}$
ν - Elasticity of substitution between sector-specific consumption	0.002
θ - Elasticity of substitution between educated and uneducated labor	1.53
η - Elasticity of substitution between female and male labor	2.27
Panel B: Data Identified Parameters	
$\{Z_A^{US}, Z_M^{US}, Z_S^{US}\}_{t=T}$ - Sector-specific technology	$\{3.78, 2.00, 1.36\}$
$\{\zeta_{EA,0}, \zeta_{NA,0}, \zeta_{EM,0}, \zeta_{NM,0}, \zeta_{ES,0}, \zeta_{NS,0}\}$ - Female productivity in 1980	$\{0.25, 0.25, 0.24, 0.29, 0.37, 0.40\}$
$\{\zeta_{EA,T}, \zeta_{NA,T}, \zeta_{EM,T}, \zeta_{NM,T}, \zeta_{ES,T}, \zeta_{NS,T}\}$ - Female productivity in 2005	$\{0.31, 0.30, 0.32, 0.31, 0.42, 0.46\}$
$\{\chi_{A,0}, \chi_{M,0}, \chi_{S,0}, \chi_{A,T}, \chi_{M,T}, \chi_{S,T}\}$ - Educated productivity	$\{0.28, 0.31, 0.45, 0.36, 0.47, 0.60\}$
$\{\alpha_0, \alpha_T\}$ - Educational assortative mating	$\{2.80, 2.16\}$
Panel C: Calibrated Parameters	
$\{\mu_m, \mu_f, \sigma_m, \sigma_f\}$ - Education cost distribution	$\{3.75, 0.34, 7.16, 1.37\}$
$\{\mu_E, \mu_N, \sigma_E, \sigma_N\}$ - Home productivity distribution	$\{-3.66, -2.80, 1.68, 0.078\}$
$\left(\frac{p_{A,0}}{Y_0} A\right), \phi_A, \phi_M$ - Consumption parameters	$\{0.024, 1.60 \times 10^{(-7)}, 0.37\}$

Note: The table reports the values and interpretations of the quantitative model parameters under the benchmark calibration to match key moments in the U.S. data.

We then directly estimate one by one some technological parameters that our model analytically identifies from the observed data moments. The data-identified parameters include the technology parameters, $\{Z_A^{US}, Z_M^{US}, Z_S^{US}\}_{t=T}$, $\{\zeta_{EA}, \zeta_{NA}, \zeta_{EM}, \zeta_{NM}, \zeta_{ES}, \zeta_{NS}\}_{t=0,T}$, $\{\chi_A, \chi_M, \chi_S\}_{t=0,T}$, and the marriage parameters, $\{\alpha_0, \alpha_T\}$. We report values of these 23 parameters in Panel B of Table 3.

First, we estimate the rise in sector-specific TFP, $\{Z_A^{US}, Z_M^{US}, Z_S^{US}\}_{t=T}$, through real labor productivity growth in each sector from the KLEMS data, following [Ngai and Petrongolo \(2017\)](#) and references therein. The resulting 2005 TFP values are, $Z_{A,T} = 3.78$, $Z_{M,T} = 2.00$ and $Z_{S,T} = 1.36$, which translate into compounded annual growth rates of 0.055 for agriculture, 0.028 for industry and 0.012 for services.¹²

Second, according to Equation (11), gender-specific productivities over time, $\{\zeta_{EA}, \zeta_{NA}, \zeta_{EM}, \zeta_{NM}, \zeta_{ES}, \zeta_{NS}\}_{t=0,T}$, are determined by 16 moments in the data: the gender wage gaps

¹²[Ngai and Petrongolo \(2017\)](#) report a differential growth rate of 1.2 percent for manufacturing relative to services from 1970 to 2006, which is consistent with the 1.6 percent we obtain from 1980 to 2005.

by education type x_j and the sectoral labor force participation ratios for educated and uneducated $\frac{H_{jk}^f}{H_{jk}^m}$ in 1980 and 2005. The estimates reported in Table 3 show that female productivity weights in services in both 1980 and 2005, $\{\zeta_{ES}, \zeta_{NS}\}_{t=0,T}$, are larger than their counterparts in agricultural and manufacturing sectors. This confirms females' comparative advantages in services, which is key for SBTC and ST to jointly pull educated women into the labor force. In addition, the fact that uneducated females have a larger ζ than educated ones suggests that uneducated females have a greater comparative advantage over uneducated males in low-skilled services, which are more substitutable to home production, as compared to educated females in skilled services.

Importantly, our model predicts that female productivity increases in all sectors from 1980 to 2005, providing evidence of GBTC as in Heathcote et al. (2010) and Cerina et al. (2021). While GBTC can be interpreted either as changes in technology or as the removal of gender-specific discrimination over time, we do not take a stand on the interpretation but follow the term as used in the literature.

Third, according to Equation (13), the six weights on sectoral educated labor input over time, $\{\chi_A, \chi_M, \chi_S\}_{t=0,T}$, are determined by the six sectoral educated wage bill shares I_{Ek} and the six sector-specific relative college-to-non-college prices π_k in 1980 and 2005. Table 3 reports the resulting estimates of χ_k , which strongly increase in all three sectors over time, suggesting that skill-biased technological change is an aggregate phenomenon in the U.S. We also find that SBTC is more pronounced in the manufacturing sector than the service sector, consistent with Buera et al. (2021, p. 18).

Fourth, we directly calculate the educational assortative matching parameter α to be 2.80 in 1980 and 2.16 in 2005 using the ratio of the actual share of couples who are both educated to the share as if the matching was random. The resulting decrease in α over time is consistent with Eika et al. (2019), who show a robust decline in educational assortative matching among the educated.¹³

Lastly, Panel C of Table 3 reports values of the 11 remaining parameters to be jointly calibrated: (i) the distribution of education cost by gender, $\{\mu_m, \mu_f, \sigma_f, \sigma_m\}$, (ii) the distribution of female home productivity by education level, $\{\mu_E, \mu_N, \sigma_E, \sigma_N\}$, and (iii) consumption weights and the non-homotheticity in preferences, $\{\phi_A, \phi_M, \underline{A}\}$. We adopt the TikTak algorithm in the simulated method of moments (SMM) calibration routine to minimize the distance between data targets and model moments to calibrate the 11 parameters with 12

¹³The benchmark estimates of α include singles in the calculation. When we restrict our sample to those who are married, α is estimated to be 2.91 in 1980 and 2.03 in 2005, which concur with the results in Eika et al. (2019).

moments concurrently.¹⁴

Even though the parameters in Panel C of Table 3 are jointly estimated, it is useful to discuss the intuition of how certain moments inform specific parameters. The distribution of education cost by gender, $\ln(a) \sim \mathcal{N}_j(\mu_g, \sigma_g^2)$, $g = f, m$, is captured by four parameters, $\{\mu_m, \mu_f, \sigma_f, \sigma_m\}$, which are largely governed by four moments in the data: the shares of educated females and males in $t = 0$ and $t = T$ or $\Gamma(a^{*f})_{t=0,T}$ and $\Gamma(a^{*m})_{t=0,T}$. The distribution of home productivity by gender, $\ln(x) \sim \mathcal{N}_j(\mu_j, \sigma_j^2)$, $j = E, N$, are captured by $\{\mu_E, \mu_N, \sigma_E, \sigma_N\}$, which are informed by the shares of women not in the labor force by education level in $t = 0$ and $t = T$. Lastly, consumption weights and the non-homothetic term in preferences, $\{\phi_A, \phi_M, \underline{A}\}$, are identified by four data moments of the initial and terminal value-added shares in manufacturing and agriculture.

As a result, our calibration implies that females have a lower average log education cost of 0.34 compared to the value of 3.75 for males, even though males have higher college-educated shares in 1980. This difference in education costs is consistent with the evidence found in Becker et al. (2010) and it contributes to the overtaking of males by females in education. The average home productivity for educated females is lower than that for uneducated females, -3.66 compared to -2.80 . This difference is partially driven by the higher LFP for educated females. For preferences, we find a non-trivial value of 0.024 for the non-homothetic term $\left(\frac{p_{A,0}}{Y_0} \underline{A}\right)$ to match the U.S. sectoral consumption shares over time. Consistent with the literature, we find a low value for the consumption weight of food and much larger weights of 0.37 and 0.63 for manufacturing and services, respectively.

Panels A in Table 4 reports the data moments directly used in identifying ζ 's and their resulting model predictions in equilibrium. Consistent with the patterns in data, the model predicts that the gender wage gap is smaller for the educated (0.70) than for the uneducated (0.66) in 1980 but the opposite is true in 2005: the gap is smaller for the uneducated (0.82) than for the educated (0.72). The model also predicts that the female to male ratio among the educated labor force is the highest in services in both years (0.67 in 1980 and 1.04 in 2005), compared to manufacturing (0.16 and 0.38) and agriculture (0.18 and 0.33), closely matching the data. Yet the model slightly underpredicts the female labor shares among the uneducated in each sector.

Panel B in Table 4 reports the data moments that directly identify χ 's and their model counterparts. The model matches the wage bill share by sector well, both in 1980 and 2005. Meanwhile, the model slightly overestimates the π_k 's, the relative college-to-non-college price

¹⁴See McFadden (1989). In addition, Arnoud et al. (2019) shows that the TikTak algorithm is the strongest performer among a series of global optimization algorithms.

Table 4: Moments in the Model vs Data

Moment	Data	Model
Panel A: Moments identifying ζ 's		
Educated gender wage gap (x_E) in $t = 0, T$	{0.69,0.70}	{0.70,0.72}
Uneducated gender wage gap (x_N) in $t = 0, T$	{0.63,0.78}	{0.66,0.82}
Educated female/male service labor ratio in $t = 0, T$	{0.69,1.12}	{0.67,1.04}
Educated female/male manufacture labor ratio in $t = 0, T$	{0.16,0.42}	{0.16,0.38}
Educated female/male agriculture labor ratio in $t = 0, T$	{0.19,0.36}	{0.18,0.33}
Uneducated female/male service labor ratio in $t = 0, T$	{1.17,1.26}	{1.04,1.11}
Uneducated female/male manufacture labor ratio in $t = 0, T$	{0.39,0.30}	{0.34,0.26}
Uneducated female/male agriculture labor ratio in $t = 0, T$	{0.25,0.26}	{0.22,0.23}
Panel B: Moments identifying χ 's		
Educated service wage bill share (I_{ES}) in $t = 0, T$	{0.38, 0.56}	{0.36, 0.54}
Educated manufacture wage bill share (I_{EM}) in $t = 0, T$	{0.20, 0.33}	{0.20, 0.33}
Educated agriculture wage bill share (I_{EA}) in $t = 0, T$	{0.17, 0.22}	{0.16, 0.20}
Relative college service price (π_S) in $t = 0, T$	{1.30, 1.96}	{1.59, 2.40}
Relative college manufacture price (π_M) in $t = 0, T$	{1.20, 2.02}	{1.48, 2.49}
Relative college agriculture price (π_A) in $t = 0, T$	{1.26, 2.03}	{1.57, 2.50}
Panel C: Jointly Target Moments in Calibration		
Male educated shares in $t = 0, T$	{0.24, 0.28}	{0.24, 0.28}
Female educated shares in $t = 0, T$	{0.16, 0.30}	{0.16, 0.30}
Educated women in LF in $t = 0, T$	{0.76, 0.81}	{0.77, 0.83}
Uneducated women in LF in $t = 0, T$	{0.61, 0.72}	{0.60, 0.71}
Manufacturing value-added share (y_M) in $t = 0, T$	{0.32, 0.22}	{0.35, 0.28}
Agricultural value-added share (y_A) in $t = 0, T$	{0.03, 0.01}	{0.04, 0.01}

Note: This table reports the moments used in the benchmark calibration and the corresponding model predictions.

by sector, in the initial year 1980 as the productivity parameters are estimated directly from the data and the benchmark model does not perfectly match gender wage gaps or employment shares. Nevertheless, the relative increases in π_S for the benchmark model almost exactly match the data, with an increase of 51 percent versus 52 percent, respectively; for π_k when $k = M, A$, they match exactly, with an increase of 68 percent in manufacturing and 60 percent in agriculture.

As shown in Panel C in Table 4, the model matches well the desired moments in the SMM

calibration routine. The model perfectly matches the education shares by gender in the data, which increases from 24% to 28% for males and from 16% to 30% for females. The model predicts that female LFPR increases from 77 percent to 82 percent among the educated and from 60 percent to 71 percent among the uneducated, closely matching the data. Meanwhile, the manufacturing and agricultural value-added shares decrease by seven and three percentage points in the model, respectively, compared to ten and two percentage points in the data.

Lastly and importantly, regarding the matching market, our model predicts that α_N is 1.11 in 1980 and 1.19 in 2005, matching well the corresponding data values of 1.10 and 1.12. This result in turn confirms that our model is robust to different types of specifications to characterize the varying assortative matching over time.

4.2 U.S. Benchmark Results

The top panel in Table 5 reports the over-time variations in the gender education gap and female LFPR by education level in the data and the model. By construction, they match well. In particular, the gender education gap narrows by 39.4 percentage points between 1980 and 2005 in the U.S. and by 39.9 percentage points in the model. Meanwhile, our model predicts that among females, university graduates and non-university graduates' LFPRs increase by 5.3 and 10.9 percentage points, respectively, matching the data moments closely.

Table 5 further reports the results of quantitative decomposition exercises using our parameterized model. It highlights the quantitative importance of ST and SBTC in explaining the narrowing gender education gap and the increasing female labor force participation rates. Row (i) of the bottom panel shows the model predictions when we allow only for ST in the model, by varying Z_k^{US} between 1980 and 2005 but keeping χ 's, ζ 's, α , and Z_h at the 1980 level. ST alone accounts for 62 percent of the benchmark model changes in the gender education gap over time, but it over-predicts the change in female LFPR. Allowing only for SBTC as in row (ii), on the other hand, contributes to only one-fifth of the narrowing gender education gap, but it decreases the female LFPR over time, contrary to the data. SBTC alone discourages women from working in the market due to the comparative advantage males enjoy in manufacturing and agricultural productions, which constitute a large share of an economy with only SBTC but not ST. This is especially true during the time period studied, when manufacturing has the most prominent SBTC. As a result, ST and SBTC are highly complementary in incentivizing women to obtain education and participate in the labor market. To further highlight this point, row (iii) reports a counterfactual of allowing

Table 5: U.S. Benchmark Decomposition between 1980 and 2005

		Female Labor Force Participation		
		Gender Edu. Gap	University	Non-University
Δ P.P. in Data		-39.4	5.0	10.6
Δ P.P. in Model		-39.9	5.3	10.9
Contributions to Empowering Women in Model				
(i)	Only ST	61.9%	170.2%	366.7%
(ii)	Only SBTC	20.6%	-210.8%	-490.0%
(iii)	Only SBTC in Services	23.8%	-149.5%	-440.0%
(iv)	ST and SBTC	116.8%	76.1%	111.1%
(v)	Only GBTC	7.4%	-28.8%	-93.6%
(vi)	Only HTC	-1.9%	-48.1%	-137.4%
(vii)	Only Assortative Matching	-7.2%	-44.9%	-157.5%

Note: The top panel of this table reports the data and the model percentage point change in the gender education gap and in the LFPRs for females who are university and non-university graduates, respectively, between 1980 and 2005 in the U.S. The bottom panel reports the model predictions in percent of the benchmark prediction when we only allow selected mechanisms to vary over time.

only for SBTC in the service sector, which accounts for about one-fourth of the narrowing gender education gap, precisely being three percentage points higher than that of the counterfactual allowing for SBTC in all sectors. The corresponding values for a counterfactual of allowing for SBTC only in manufacturing or agriculture are two percent and minus two percent, respectively (not shown in Table 5). This result confirms that SBTC in services is the only sectoral SBTC that matters for the narrowing gender education gaps. By combining SBTC and ST in row (iv), the model induces more women than men to become educated and correctly predicts that the female LFPR increases less for the university-educated than the uneducated, despite slightly over-predicting the the magnitude of the difference.

In contrast to the crucial role of the complementarity between SBTC and ST in reversing the gender education gap, rows (v) to (vii) in Table 5 report the model's predictions when we only allow for one of the other three mechanisms. Only allowing for GBTC contributes modestly (7.4%) to the benchmark prediction on the gender education gap. In this case, even though the gender wage gap narrows, the general equilibrium effects of rising consumption prices

outweigh the relative increase in female wages. As a result, women, particularly uneducated women, shift from market to home production. Only allowing for HTC, which increases home productivity marginally, widens the gender education gap because it incentivises women to stay at home rather than work in the market. Lastly, only reducing the educated assortative matching parameter α widens the 1980 gender education gap by seven percent. Adding all these three mechanisms with SBTC and ST matches the model’s quantitative predictions with the data. We conclude that SBTC, especially in services, combined with ST are the main drivers of the narrowing gender education gap in the U.S.

4.3 Model Validation Using Other High-Income Countries

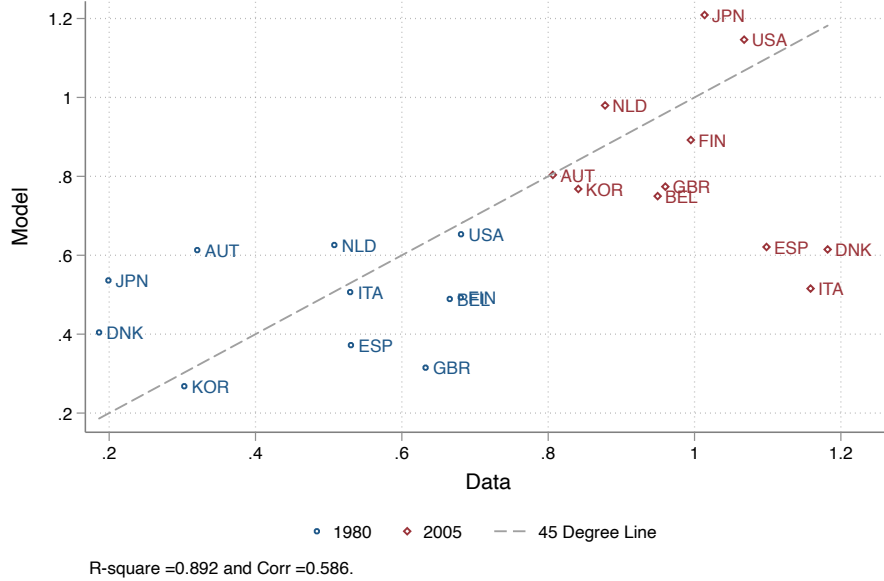
In this section, we validate the model by assessing its performance of predicting the *non-targeted* variations in the gender education gap over time and across high-income economies from the KLEMS data. We start with the benchmark parameter values but re-calculate only the data-identified parameters Z ’s, χ ’s, and ζ ’s for each country-year sample using the same estimation procedure as for the U.S.¹⁵ We further calibrate the home productivity parameter Z_h to match the female LFPR in 1980 and 2005. Data moments of the LFPR are taken from OECD.Stat covering all 25–54 year old women. Appendix Table A2 reports the estimates that we use for the exercise. Figure A4 plots the female LFPR in the data and in the model for KLEMS countries in 1980 and 2005.

Figure 5 plots the model’s predictions of the gender education gaps against the data, where data moments are computed using the educated population size by sex and age groups from Barro and Lee (2013). The R-square shows that our model accounts for 89 percent of the variation in the gender education gap over time across KLEMS countries. The correlation between the education gaps in the data and the model is 0.59.

Figure 6 shows the percent of the change in the gender education gap between 1980 and 2005 that can be explained by the mechanisms in our model for each KLEMS country. The first set of columns shows that SBTC and ST account well for the evolution in half of the countries including Belgium, Finland, the UK, Japan, and Korea, with an average explanatory power of 90 percent. The second and third sets of columns present the effect of first adding country-specific GBTC and further adding HTC, respectively. Overall, GBTC and HTC have smaller effects on narrowing gender education gaps, with the exception of

¹⁵Ideally, we would also like to use the actual estimated assortative matching parameter for each country-year sample. Unfortunately, this moment is not available in the KLEMS data; very few desired samples have household surveys where α can be identified in both time periods. Hence, we use the estimated α for the U.S. in 2005.

Figure 5: Model Predictions on the Gender Education Gap Across KLEMS Countries



Note: This figure plots the gender education gap in the base model against that in the data. It also reports the R-square of regressing one on the other and the correlation coefficient between the two.

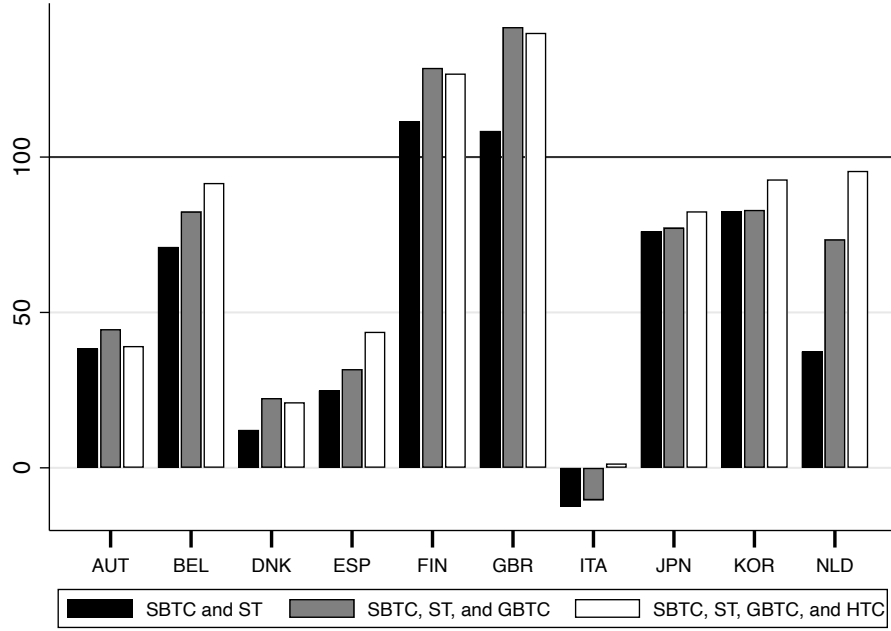
the Netherlands. In addition, our three mechanisms explain less than half of the patterns in Austria, Denmark, Spain, and Italy. For the latter set of idiosyncratic countries, there must be mechanisms other than our proposed ones that push down the gender education gap over time. For example, such mechanisms could include changing social norms or country-specific policies and reforms.¹⁶ We conclude that the complementarity between SBTC and ST remains the crucial mechanism for the model's quantitative power in explaining the narrowing gender education gap broadly in KLEMS countries.

5 Model Predictions Across Income Levels

We have developed and validated a model of the gender education gap that matches the key characteristics of the advanced economies from 1980 to 2005. In this section, we assess the model's predictions on the gender education gap over the full range of the world income distribution. To do so, we fix all the parameters but allow for exogenous variations in three dimensions: (i) sectoral productivity $\{Z_A, Z_M, Z_S\}$, (ii) home productivity Z_h , (iii) sectoral

¹⁶Specifically, Denmark is an outlier for the particularly large decrease in the gender education gap, with the ratio of female to male educated share increasing by one, compared to the average value of 0.5. It also experienced relatively small SBTC. Italy is also unique in the sense that service productivity decreased rather than increased from 1980 to 2005. Spain is a mix of the experience of these two countries.

Figure 6: Share of the Change in the Gender Education Gap Explained by the Model



Note: This figure plots the share of the gender education gap in each KLEMS country that is explained by i) a model with only SBTC and ST, ii) an alternative model additionally allowing for GBTC, and iii) another alternative model additionally allowing both for GBTC and HTC.

skill intensity $\{\chi_A, \chi_M, \chi_S\}$ in the three market sectors, and (iv) educational assortative matching α in the marriage market, to match key features of labor markets in low-, middle-, and high-income countries. Due to the lack of available data on the gender wage gap to identify the gender weights in the production function for developing economies, we fix those parameters (ζ 's) at the U.S. level in 1980 for model predictions across income levels in the main analysis. We also present the model predictions when we calibrate the model while fixing ζ 's at the U.S. 2005 level. We then test the model predictions on the gender education gap across development levels.

As we consider the full spectrum of world income distribution, it is crucial to match women's labor market characteristics, particularly the high LFPRs and large agricultural sectors that distinguish low-income countries from the high-income economies we have discussed thus far. Therefore, we target only female moments. As the moments for direct identification are not available in developing economies, we calibrate the economy-wide parameters $\{Z_A, Z_M, Z_S\}$, Z_h , and $\{\chi_A, \chi_M, \chi_S\}$ to match the median values of the seven female moments: (i) LFPR by education, (ii) employment shares of agriculture and services, and (iii) shares of educated workers in each sector. The assortative matching parameter is directly identified by the

shares of different types of households by education level in the data as before.

Table 6: Parameter Values by Income Tercile

Calibrated Parameter	Value
$\{Z_A, Z_M, Z_S\}_{high}$	$\{0.48, 1.05, 0.84\}$
$\{Z_A, Z_M, Z_S\}_{middle}$	$\{0.18, 0.76, 0.54\}$
$\{Z_A, Z_M, Z_S\}_{low}$	$\{0.050, 0.76, 0.42\}$
$\{\chi_A, \chi_M, \chi_S\}_{high}$	$\{0.24, 0.43, 0.60\}$
$\{\chi_A, \chi_M, \chi_S\}_{middle}$	$\{0.14, 0.42, 0.35\}$
$\{\chi_A, \chi_M, \chi_S\}_{low}$	$\{0.080, 0.21, 0.30\}$
$\{Z_h\}_{high,middle,low}$	$\{0.40, 0.34, 0.34\}$
Estimated Parameter	Value
$\{\alpha\}_{high,middle,low}$	$\{3.96, 7.27, 16.63\}$

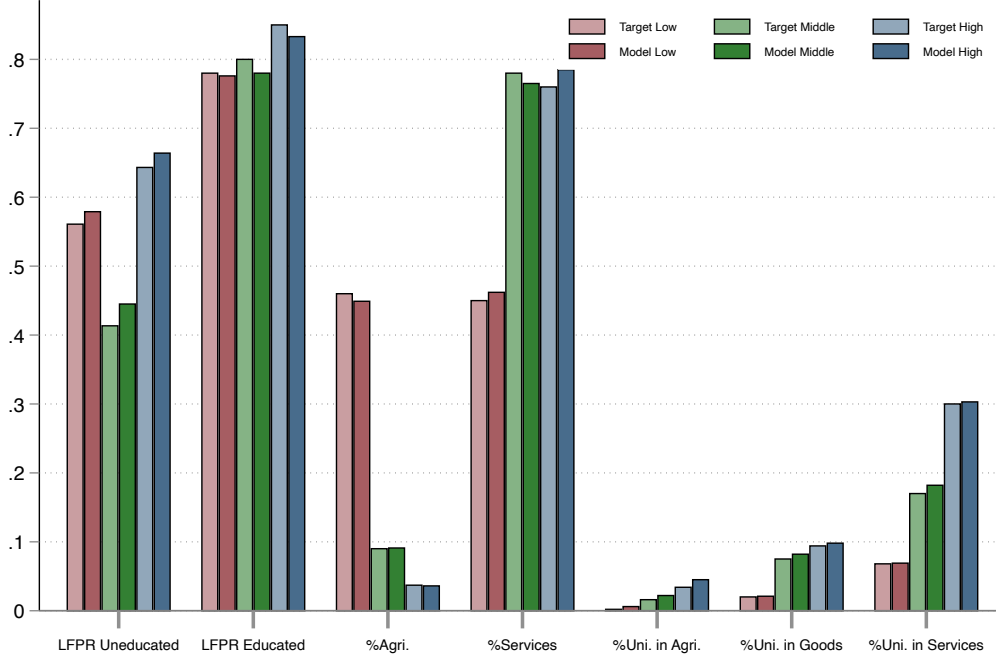
Note: The table reports the values of the parameters varying across the median low-, middle-, and high-income countries.

Table 6 presents the values of the calibrated technological parameters and the estimated assortative matching parameter in high-, middle-, and low-income countries. Consistent with the literature, we find that productivity growth is the highest in agriculture and lower in manufacturing and services, which implies ST across countries. Meanwhile, we also find evidence of SBTC within all three sectors across countries, as indicated by larger estimates of χ 's in more-developed economies. Our model suggests that the home productivity does not change between a median low- to a middle-income country, but increases significantly for a median high-income country. In addition, while α decreases with development, the magnitudes of decline across countries are much bigger than that in the U.S. over time.

Figure 7 plots the moments targeted in the data and model. Our calibrated model closely matches the U-shaped LFPR for uneducated females and the increasing LFPR for educated females, declining agriculture employment among females, and rising services employment among females across countries. For the share of educated labor in each sector's female labor force, our model also closely matches the increasing skill intensity among female workers in all three sectors. In addition, our cross-country model predicts that α_N , the assortative matching parameter between uneducated women and men, increases from 1.1 in a low-income country to 1.3 in middle- and high-income countries, which is broadly consistent with the trends and values shown in Figure A2.

We further test whether or not the model's predictions of the *non-targeted* female intensity by sector and education level are consistent with the empirical pattern. Table 7 reports the

Figure 7: Targeted Moments in the Model and the Data by Income Tercile



Note: This figure plots the moments targeted in the cross-country calibration and the corresponding model predictions in the median low-, middle-, and high-income countries.

Table 7: The FLFS Difference between High- and Low-Income Country

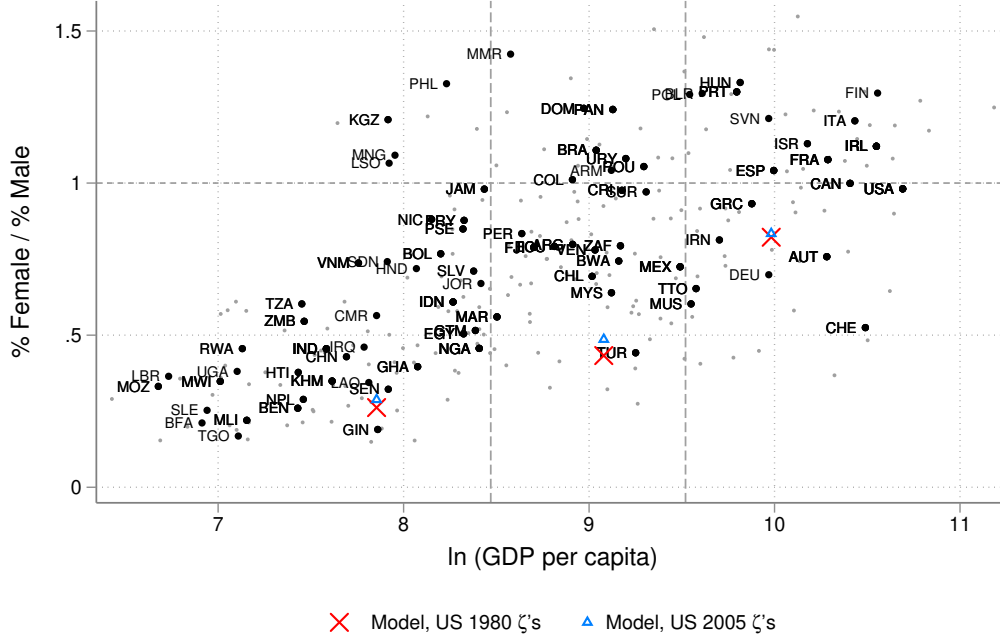
	Aggregate		Non-University		University	
	Data	Model	Data	Model	Data	Model
Agriculture	-5.5	-7.9	-6.1	-8.4	11.6	11.7
Manufacturing	-1.1	-8.1	-1.5	-10.2	10.2	10.5
Services	9.1	-6.1	9.4	-11.2	22.6	24.0

Note: This table reports the percentage point value of the female labor force share in the median high-income country less that in the median low-income country. We report this statistic in the aggregate and by education level for each sector in the model and in the data.

difference between high- and low-income countries in the FLFS by sector in the aggregate and by education level in the data and the model. Without targeting the moments, our model correctly predicts the aggregate decline in the FLFS in agriculture and manufacturing, although it fails to generate the aggregate increase of the FLFS in services. Furthermore, consistent with the data, the model correctly predicts that the aggregate patterns are driven by uneducated workers in all three sectors. Most importantly, the model predicts that the FLFS among educated workers rises by 12, 11, and 24 percentage points in agriculture,

manufacturing, and services, respectively, moving from low- to high-income countries, closely matching the trend and magnitudes of 12, 10, and 23 percentage points in the data.

Figure 8: Gender Education Gap in the Model and the Data



Note: This figure plots the gender education gap against log GDP per capita. Each black dot represents one country as in Panel (a) of Figure 1; red crosses and blue triangles are the predictions of the quantitative model fixing gender-specific productivities at the U.S. level in 1980 and 2005, respectively.

To what extent does our model explain the narrowing gender education gap with development across countries? As plotted in Figure 8 in red crosses, our model under-predicts that females are 82 percent as likely as males to obtain a college degree in the median high-income country, compared to the data moment of 102 percent. The model also predicts that the gender education gap widens by 39 percentage points when moving from the median high-income country to the median middle-income country, and by 17 percentage points when further moving to the median low-income country. In the data, the corresponding differences are 21 and 35 percentage points. Although our model over-predicts the levels of the gender education gap, especially in middle-income countries, it perfectly matches the difference of 56 percentage points between low- and high-income countries. Model predictions when we fix the values of ζ 's at the U.S. 2005 level are plotted in blue triangles in Figure 8, which are almost identical to the baseline results.

To conduct the decomposition analysis between high- and middle-income countries, we allow only the selected mechanism to vary in a median middle-income country while fixing other

Table 8: Decomposition of the Mechanisms of Cross-country Predictions

Difference	Female LFPR		Gender Edu. Gap	
	High – Middle	Middle – Low	High – Middle	Middle – Low
	(1)	(2)	(3)	(4)
Data P.P.	25.4	-13.2	21.0	34.9
Benchmark P.P.	22.5	-10.8	38.9	17.2
Contributions to Empowering Women in Model				
Only ST	253.8%	669.0%	87.9%	101.9%
Only SBTC	-125.3%	8.9%	51.2%	88.0%
Only SBTC services	-124.0%	7.0%	42.4%	69.6%
Only ST and SBTC	169.3%	123.6%	106.7%	127.0%
Only ST and SBTC services	194.2%	564.3%	105.4%	52.2%
Only HTC	-80.0%	0.0	-2.8%	0.0

Note: The top panel of this table reports the percentage point difference of the female LFPR and the gender education gap in the data and the model between the median high- and middle-income country and between the median middle- and low-income country. The bottom panel reports the model predictions in percent of the benchmark prediction when we only allow selected mechanisms to vary across countries.

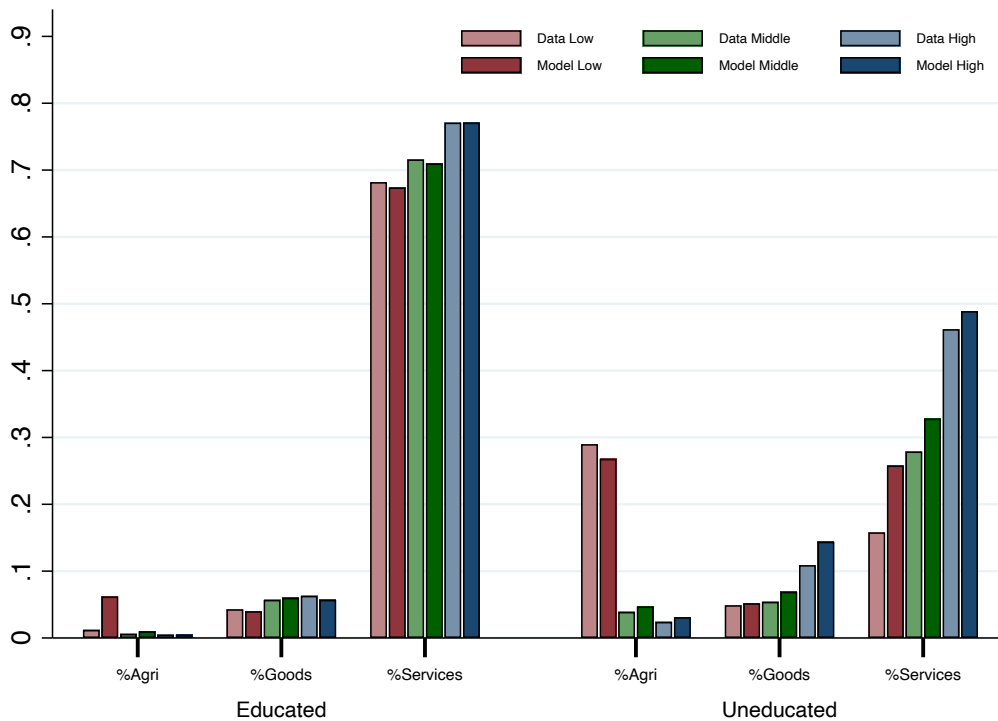
parameters at the level of a median high-income country for model predictions. Table 8 reports the corresponding results in Columns (1) and (3). Consistent with our findings for the U.S. over time, the difference in the gender education gap between high- and middle-income countries in the model is jointly explained by ST and SBTC (107 percent), leaving a minor negative effect that is accounted for by other mechanisms. Meanwhile, just as for the time series pattern in the U.S., the difference in female LFPRs is solely explained by ST, while SBTC and HTC predict a contrasting pattern to the data.

We then conduct the decomposition analysis between middle- and low-income countries and report the results in Columns (2) and (4) of Table 8. Either ST or SBTC alone contributes significantly to the narrowing gender education gap in the model and the joint explanatory power is 127 percent of the benchmark prediction. As LFPRs are high in low-income countries, a rise in ST to the median middle-income country level reduces rather than raises female LFPR, in contrast to the difference between high- and middle-income countries and U.S. time series patterns. Instead, with ST alone, women exit the labor market as they do not need to contribute to subsistence consumption. SBTC alone makes a limited contribution to the fall in the LFPR in low-income countries because men still have a strong comparative advantage in the labor market. Thus, only the combination of SBTC and ST can explain the initial relative fall and subsequent rise (U-shaped pattern) of the female LFPR.

Only allowing HTC decreases the female LFPR from high- to middle-income countries be-

cause of a bigger opportunity cost of females working in the market, contrary to the data. The estimated HTC between low- and middle-income countries is too small to have meaningful effects on the female LFPR. Regarding the narrowing gender education gap across income levels, HTC consistently has little explanatory power. Only varying assortative matching accounts for literally zero of the benchmark predictions across countries (not listed in Table 8), because when α is too large (bigger than three), the household share of both educated agents is constrained by the share of educated females but not affected at all by changes in α . Therefore, we conclude that the complementarity between SBTC and ST is the single most important factor in explaining the narrowing gender education gap with economic development.

Figure 9: Non-targeted Educated and Uneducated Female Employment by Sector



Note: This figure plots the uneducated and educated female employment shares by sector in the data and the model for the median low-, middle-, and high-income countries.

Role of Marketization of Home Services. What is the role of endogenous marketization – the shift towards the production of personal services in the market rather than at home – in shaping the narrowing gender education gap across countries? Figure 9 reports the *non-targeted* educated and uneducated females’ employment shares in agriculture, manufacturing, and services across income levels, showing a good match between the data and the model. Consistent with the literature focusing on the U.S. over time (see e.g., [Ngai and](#)

Petrongolo, 2017; Zhang, 2023), we find a surge in low-skill women’s employment in services from the median middle-income country to the median high-income country, increasing from 33 percent in the model (and 28 percent in the data) to 49 percent in the model (and 46 percent in the data). Meanwhile, the increase in high-skill women’s employment in services is much smaller, at six percentage points, both in the data and the model. These results suggest a strong pattern of marketization of home production when comparing middle- and high-income countries.

However, the comparison between the median low- and middle-income countries paints a different picture. In the model, low-skill women’s employment in services only increases by 7 percentage points (or 12 percentage points in the data), which is much smaller than the 16 percentage point surge (or 18 percentage points in the data) observed between middle- and high-income countries. This difference arises because, while SBTC and ST attract high-skill women into market work in low-income countries, improvements in agricultural productivity help households meet subsistence needs, thereby reducing the LFPR of low-skill women. Consequently, the limited marketization of home production constrains the extent to which the gender education gap narrows with economic development in low- and middle-income countries.

6 Conclusions

In this paper, we focus on understanding the gender education gap across countries both empirically and theoretically. We start by documenting the strong negative relationship between income and the gender education gap. The robust relationship is particularly impressive given the enormous heterogeneity in social and cultural norms across countries; for example, girls are attacked in many countries for getting educated (O’Malley, 2010), while the Nordic countries have inclusive education.

To investigate the causes of the sharp decrease in the gender education gap with economic development, we develop a three-sector model featuring skilled-biased technological change, structural transformation, endogenous education and female labor supply decisions, and assortative household formation. We validate the model by testing its predictions on a large set of out-of-sample developed countries. Additionally, the model decently matches two sets of *non-targeted* moments: i) differences in FLFS between high- and low-income countries in the aggregate and by education level, and ii) the educated and uneducated female employment by sector across countries.

Through the lens of our quantitative model, females have a comparative advantage in services, and structural transformation towards services *complemented* by SBTC in services is the single most important factor driving the narrowing gender education gap with economic development. In contrast, SBTC in sectors where women do not hold a comparative advantage leads to a slight increase in the gender education gap. We also show that GBTC, HTC, and varying assortative matching play at most a modest role in explaining the robust trend of the decrease in the gender education gap with development, although these other mechanisms have some merit in explaining cross-country variations conditional on income levels.

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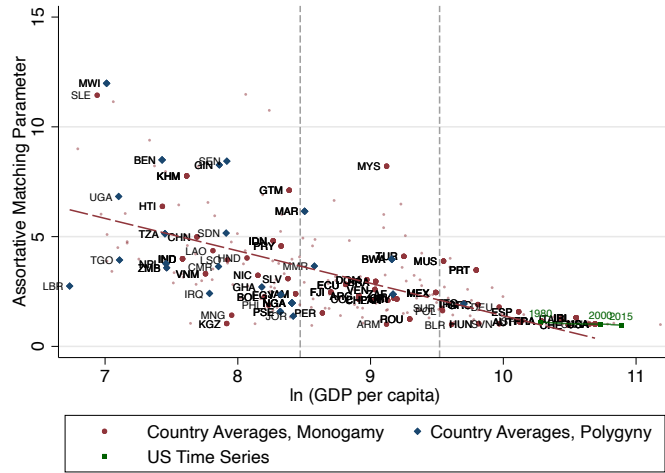
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Online Appendices

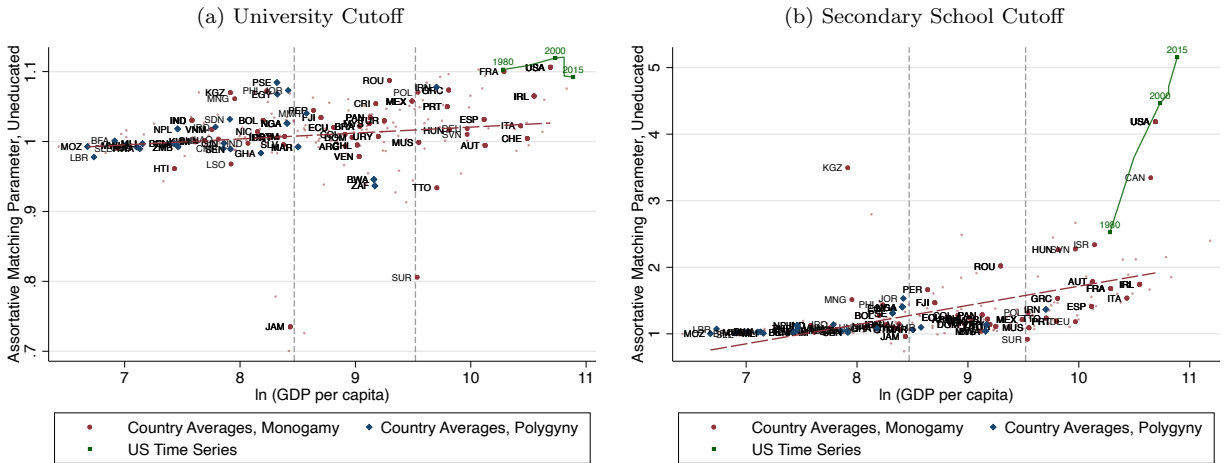
A Additional Figures and Tables

Figure A1: Educational Assortative Matching, Secondary School Threshold



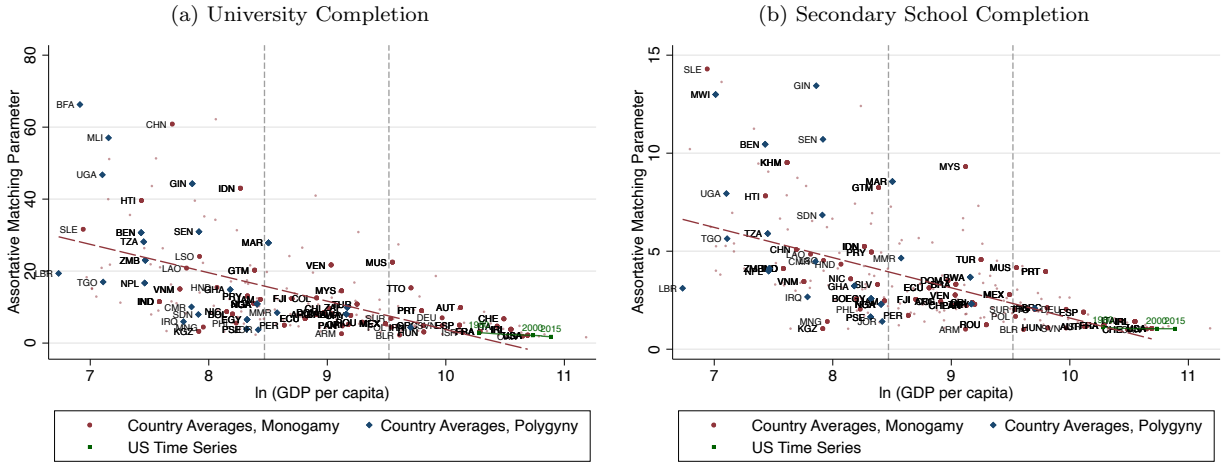
Note: This figure plots α values against log GDP per capita for each country-year sample (light red dot) and country averages (blue diamond for countries that have polygamous unions and dark red dot for the rest). Observations with secondary-school α values larger than the country-average 95th percentile value of 11.98 are dropped in this figure.

Figure A2: Educational Assortative Matching among Uneducated



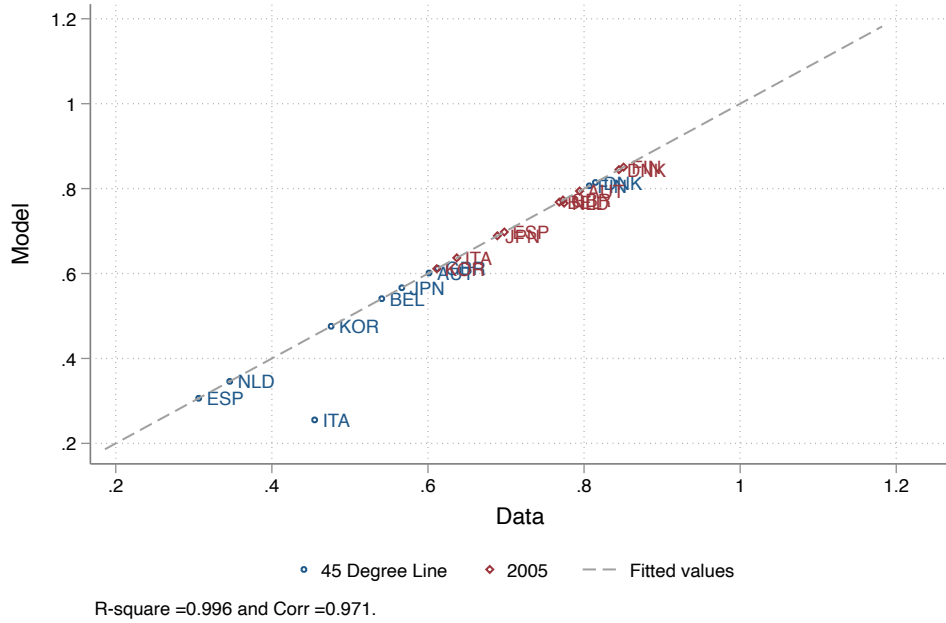
Note: This figure plots α_N values against log GDP per capita for each country-year sample (light red dot) and country averages (blue diamond for countries that have polygynous unions and dark red dot for the rest). Observations larger than the country-average 95th percentile are dropped.

Figure A3: Educational Assortative Matching, Married Only



Note: Restricting each sample to married households, this figure plots α values against log GDP per capita for each country-year sample (light red dot) and country averages (blue diamond for countries that have polygynous unions and dark red dot for the rest). Observations larger than the country-average 95th percentile are dropped.

Figure A4: Model Predictions of Female Labor Force Participation Rate



Note: This figure plots the female LFPR predicted by the model against that in the data in all KELMS samples using data-identified Z 's, χ 's, ζ 's, and calibrated Z_h 's.

Table A1: Slope of Female Labor Force Share within Sector (%) on log GDP per capita

	Aggregate	By Education Group	
		Non-University	University
Agriculture	-5.39*** (1.46)	-5.50*** (1.46)	4.25*** (0.98)
Manufacturing	-0.46 (.84)	-0.73 (.86)	3.81*** (.74)
Services	3.53*** (1.07)	3.19** (1.15)	8.07*** (1.00)
Obs.	83	83	83

Note: ***, ** and * indicate statistical significance at the one-percent, five-percent and 10-percent levels. Standard errors are shown in parentheses. We use the industry variable (*indgen*) for classification. The agricultural sector includes agriculture, fishing, and forestry; the manufacturing sector includes mining, extraction, manufacturing, construction, and utilities; the service sector includes wholesale and retail trade, hotels, restaurants, transportation, storage, communications, financial services, insurance, public administration, defense, services not specified, business services, real estate, education, health, social work, other services, and private household services.

Table A2: Calibrated Parameters for KLEMS Countries

Country	$\{Z_A, Z_M, Z_S\}_{1980}$	$\{Z_A, Z_M, Z_S\}_{2005}$	$\{\chi_A, \chi_M, \chi_S\}_{1980}$	$\{\chi_A, \chi_M, \chi_S\}_{2005}$
AUT	1.15,0.82,0.93	2.5,1.97,1.11	0.26,0.15,0.04	0.36,0.22,0.19
BEL	0.92,0.77,0.96	2.23,1.7,1.14	0.31,0.2,0.09	0.39,0.32,0.16
DNK	0.59,0.72,0.8	2.54,1.14,1.06	0.23,0.11,0.13	0.3,0.18,0.1
ESP	0.3,0.48,0.55	1,0.78,0.61	0.41,0.2,0.06	0.54,0.35,0.24
FIN	0.85,0.54,0.68	2.22,1.72,0.87	0.48,0.36,0.37	0.59,0.46,0.46
ITA	0.38,0.46,0.59	1.36,0.7,0.53	0.24,0.09,0.04	0.39,0.12,0.12
JPN	1.08,1.08,1.1	2.27,2.66,1.9	0.4,0.31,0.25	0.52,0.43,0.37
KOR	0.23,0.24,0.43	1.11,1.62,0.67	0.58,0.46,0.45	0.66,0.55,0.6
NLD	1.01,1.17,1.39	2.22,1.85,1.75	0.29,0.13,0.02	0.37,0.25,0.29
UK	0.58,0.49,0.68	1.3,1.38,0.98	0.31,0.18,0.15	0.48,0.35,0.26
Country	$\{\zeta_{EA}, \zeta_{NA}, \zeta_{EM}, \zeta_{NM}, \zeta_{ES}, \zeta_{NS}\}_{1980}$	$\{\zeta_{EA}, \zeta_{NA}, \zeta_{EM}, \zeta_{NM}, \zeta_{ES}, \zeta_{NS}\}_{2005}$	Z_{h1980}	Z_{h2005}
AUT	0.37,0.5,0.31,0.38,0.48,0.48	0.45,0.47,0.37,0.36,0.5,0.48	1.74	1.99
BEL	0.28,0.34,0.2,0.3,0.37,0.43	0.34,0.35,0.31,0.31,0.41,0.49	1.5	1.73
DNK	0.24,0.31,0.23,0.31,0.33,0.47	0.37,0.31,0.38,0.35,0.42,0.51	1.34	1.94
ESP	0.19,0.31,0.19,0.21,0.37,0.38	0.36,0.32,0.35,0.27,0.48,0.44	0.43	0.25
FIN	0.21,0.29,0.27,0.37,0.42,0.5	0.31,0.34,0.34,0.3,0.49,0.49	0.31	0.43
ITA	0.32,0.41,0.27,0.36,0.47,0.45	0.52,0.5,0.46,0.4,0.56,0.54	0.75	0.4
JPN	0.14,0.27,0.12,0.27,0.24,0.3	0.25,0.36,0.18,0.3,0.28,0.41	1.32	3.17
KOR	0.26,0.16,0.13,0.24,0.28,0.19	0.3,0.25,0.23,0.31,0.37,0.33	0.06	0.17
NLD	0.18,0.25,0.15,0.22,0.26,0.35	0.31,0.31,0.34,0.31,0.4,0.46	4.05	4.43
UK	0.05,0.24,0.17,0.25,0.28,0.35	0.34,0.26,0.33,0.27,0.4,0.41	0.63	0.86

Note: The table reports the estimates of the sectoral productivity, sectoral skill-intensity parameters, gender-specific productivity, and home productivity in each available KLEMS country in 1980 and 2005.

B Proofs of the theoretical results

B.1 Consumption

Given market income agents maximize market consumption. We can ignore the home production choice and solve consumption conditional on income, $y_{Wjj'}$. Substituting the budget constraint for c_S , $c_S = \frac{y_{Wjj'} - p_A c_A - p_M c_M}{p_S}$, FOCs are:

$$\frac{\partial c}{\partial c_A} = c^{\frac{1}{\nu}} \left[\phi_A^{\frac{1}{\nu}} (c_A - \underline{A})^{\frac{-1}{\nu}} - \phi_S^{\frac{1}{\nu}} (\hat{c}_S)^{\frac{-1}{\nu}} \frac{p_A}{p_S} \right] = 0 \quad (17)$$

$$\frac{\partial c}{\partial c_M} = c^{\frac{1}{\nu}} \left[\phi_M^{\frac{1}{\nu}} (c_M)^{\frac{-1}{\nu}} - \phi_S^{\frac{1}{\nu}} (\hat{c}_S)^{\frac{-1}{\nu}} \frac{p_M}{p_S} \right] = 0 \quad (18)$$

Using the two first-order conditions and the budget constraint we can derive the following relative consumption choices and consumption of market services with the assumption that home production and market services are perfect substitutes:

$$\frac{c_H + c_S}{c_A - \underline{A}} = \left(\frac{\phi_S}{\phi_A} \right) \left(\frac{p_A}{p_S} \right)^{\nu}, \quad (19)$$

$$\frac{c_H + c_S}{c_M} = \left(\frac{\phi_S}{\phi_M} \right) \left(\frac{p_M}{p_S} \right)^{\nu} \quad (20)$$

$$c_S = \frac{y_{jj'W} - p_A \underline{A} - c_H \left(p_A \left(\frac{\phi_A}{\phi_S} \right) \left(\frac{p_S}{p_A} \right)^{\nu} + p_M \left(\frac{\phi_M}{\phi_S} \right) \left(\frac{p_S}{p_M} \right)^{\nu} \right)}{p_S + p_A \left(\frac{\phi_A}{\phi_S} \right) \left(\frac{p_S}{p_A} \right)^{\nu} + p_M \left(\frac{\phi_M}{\phi_S} \right) \left(\frac{p_S}{p_M} \right)^{\nu}}. \quad (21)$$

Note that relative consumption, Equations (19) and (20), do not depend on education-sector status. Given $\Omega = \phi_S p_S^{1-\nu} + \phi_A p_A^{1-\nu} + \phi_M p_M^{1-\nu}$ and Equations (19)-(21), total consumption (the composite of market and home consumption) is

$$c = (y_{Wjj'} - p_A \underline{A} + p_S c_H) \Omega^{\frac{1}{\nu-1}}. \quad (22)$$

B.2 Home production as imperfect substitutes for services

In this section, we present a model where home production is an imperfect substitute for service consumption. The household's aggregate market and home consumption is a CES of

$$c = \left(\phi_M c_M^{\frac{\nu-1}{\nu}} + \phi_S \hat{c}_S^{\frac{\nu-1}{\nu}} + \phi_A (c_A - \underline{A})^{\frac{\nu-1}{\nu}} \right)^{\frac{\nu}{\nu-1}} \quad (23)$$

where c_M and c_A are sector-specific consumption and $\underline{A}$ is a constant as in the benchmark model. But $\hat{c}_S$ is a composite service consumption made up of home and market consumption. Given the discrete employment choices, service consumption is

$$\hat{c}_S = \left(c_H^{\frac{\sigma-1}{\sigma}} + c_S^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}. \quad (24)$$

Households maximize utility:

$$\max_{c_M, c_S, c_A, c_H(l_{jj}^f)} u(c) \quad (25)$$

s.t.

$$\sum_{i=G,S,B} p_i c_i \leq w_j^m + w_{j'}^f l_{jj'}^f \equiv y_{jj'} w. \quad (26)$$

Given market income, households maximize market consumption c . Substituting the budget constraint for c_S , $c_S = \frac{y_j - p_A c_A - p_M c_M}{p_S}$, FOCs are:

$$\frac{\partial c}{\partial c_A} = c^{\frac{1}{\nu}} \left[\phi_A (c_A - \underline{A})^{\frac{-1}{\nu}} - \phi_S (\hat{c}_S)^{\frac{\nu-\sigma}{\sigma\nu}} c_S^{\frac{-1}{\sigma}} \frac{p_A}{p_S} \right] = 0, \quad (27)$$

$$\frac{\partial c}{\partial c_M} = c^{\frac{1}{\nu}} \left[\phi_M (c_M)^{\frac{-1}{\nu}} - \phi_S (\hat{c}_S)^{\frac{\nu-\sigma}{\sigma\nu}} c_S^{\frac{-1}{\sigma}} \frac{p_M}{p_S} \right] = 0. \quad (28)$$

Then

$$c_A = \left(\frac{\phi_A p_S}{\phi_S p_A} \right)^{\nu} \left(\left(\frac{c_H}{c_S} \right)^{\frac{\sigma-1}{\sigma}} + 1 \right)^{\frac{\sigma-\nu}{\sigma-1}} c_S + \underline{A}, \quad (29)$$

$$c_M = \left(\frac{\phi_M p_S}{\phi_S p_M} \right)^{\nu} \left(\left(\frac{c_H}{c_S} \right)^{\frac{\sigma-1}{\sigma}} + 1 \right)^{\frac{\sigma-\nu}{\sigma-1}} c_S, \quad (30)$$

$$p_S c_S + \left[p_A \left(\frac{\phi_A p_S}{\phi_S p_A} \right)^{\nu} + p_M \left(\frac{\phi_M p_S}{\phi_S p_M} \right)^{\nu} \right] \left(\left(\frac{c_H}{c_S} \right)^{\frac{\sigma-1}{\sigma}} + 1 \right)^{\frac{\sigma-\nu}{\sigma-1}} c_S = y_j - p_A \underline{A}. \quad (31)$$

Equation (31) solves for c_S and Equations (29) and (30) solve for the remaining consumption allocation. A female enters the labor market if $w_j^m < p_A \underline{A}$ or if $u(c|c_H = \psi_j x) \geq u(c|c_H = x)$.

B.3 Identifications of χ_k

Given p_{Ek} and p_{Nk} as the education-specific factor prices by sector, the firms' profit maximization problem is equivalent to

$$\begin{aligned} \max_{\{H_{jk}\}} \quad & p_k Y_k - \sum_j p_{jk} H_{jk}, \text{ and} \\ \max_{\{H_{jk}^g\}} \quad & p_{jk} H_{jk} - \sum_{jg} w_{jk}^g H_{jk}^g, \text{ for } k = A, M, S, \end{aligned}$$

which solves

$$p_{Ek} = p_k Z_k H_k^{\frac{1}{\theta}} \chi_k (H_{Ek})^{-\frac{1}{\theta}}, \quad (32)$$

$$p_{Nk} = p_k Z_k H_k^{\frac{1}{\theta}} (1 - \chi_k) (H_{Nk})^{-\frac{1}{\theta}}, \text{ for } k = A, M, S. \quad (33)$$

Then the sector-specific college premium by sector π_k can be written as

$$\pi_k \equiv \frac{p_{Ek}}{p_{Nk}} = \frac{\chi_k}{1 - \chi_k} \left(\frac{H_{Ek}}{H_{Nk}} \right)^{-\frac{1}{\theta}}. \quad (34)$$

Recall that the wage bill shares by education type are defined as:

$$I_{Ek} = \frac{p_{Ek} H_{Ek}}{p_{Ek} H_{Ek} + p_{Nk} H_{Nk}}, \quad (35)$$

$$I_{Nk} = 1 - I_{Ek}, \text{ where } k = A, M, S. \quad (36)$$

Using Equation (34), the wage bill shares can be written as functions of college premium that

$$I_{Ek} = \left[1 + \pi_k^{\theta-1} \left(\frac{1 - \chi_k}{\chi_k} \right)^\theta \right]^{-1}, \quad (37)$$

which identifies χ_k for a given π_k as in Equation (13).